

claude-windows / 2139f383-5578-4e22-831e-8859ad4f69f2

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\2139f383-5578-4e22-831e-8859ad4f69f2.jsonl`
- SHA-256: `2e1c23d45866cb98128c2c541e87258d5a6a4e5842979f64a98d338163fc8c54`
- Source modified: `2026-06-10T10:34:05+00:00`
- Imported at: `2026-07-05T16:48:17+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `2139f383-5578-4e22-831e-8859ad4f69f2`
```

Transcript

1. user (2026-06-10T05:49:28.290Z)

In C:\Users\avidu\Projects\badminton-highlight-indexer, commit 7fbbbe0 ("feat(logging): standardize to logging in hot paths", 2026-06-07, on master) converted print() calls to logger calls in backend/pipeline/stitcher/compiler.py but dropped the f-string quotes on 7 lines: 100, 104, 156, 160, 181, 191, 213. Example (line 100): `logger.warning({segment\_label}: start\_time is negative ({start:.2f}s). Clamping to 0.)` ? should be `logger.warning(f"{segment\_label}: start\_time is negative ({start:.2f}s). Clamping to 0.")`. This is a SyntaxError: `python -m py\_compile backend/pipeline/stitcher/compiler.py` fails, and since backend/main.py imports VideoCompiler at module level, `import backend.main` (the FastAPI server and /api/compile) is broken on master. The 254-test suite stays green because no test imports backend.main or the compiler.

Task: (1) restore the f-string quotes on all 7 lines (keep message text identical; also check line 168 which has a mangled `f"\nSUMMARY]"` missing its opening `[`); (2) verify with `python -m compileall backend` and `python -c "import backend.main"`; (3) add a minimal regression guard test (e.g. tests/test_import_smoke.py that does compileall over backend/ or imports backend.main with heavy deps mocked) so a future syntax error can't ship green again; (4) run `python run\_all\_tests.py` (offline, mocked). Branch off master, one PR per the repo's CLAUDE.md conventions; descriptive commit message. Do not refactor anything else in the file.

2. assistant (2026-06-10T05:49:36.209Z)

I'll fix the broken f-strings in the compiler, verify imports, add a regression smoke test, and put it up as one PR off master. Starting by getting onto master and reading the damaged file.

3. user (2026-06-10T05:49:40.470Z)

```
Your branch is up to date with 'origin/master'.
Switched to branch 'master'
From https://github.com/avidullu/badminton-highlight-indexer
  4d04a73..a5477c2  master    -> origin/master
Updating 4d04a73..a5477c2
Fast-forward
 docs/NEXT_STEPS.md | 46 ++++++++++++++++++++++++++++++++++++++
 1 file changed, 46 insertions(+)
 create mode 100644 docs/NEXT_STEPS.md
```

4. user (2026-06-10T05:49:40.490Z)

```
1     import os
2     import subprocess
3     import tempfile
4     import logging
5     from typing import List, Tuple, Union
6
7     logger = logging.getLogger(__name__)
```

```

8
9 class VideoCompiler:
10     def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =
0.5):
11         self.output_dir = output_dir
12         self.end_padding = end_padding
13         self.begin_padding = begin_padding
14         os.makedirs(output_dir, exist_ok=True)
15
16     @staticmethod
17     def _parse_time(val: Union[int, float, str]) -> float:
18         """Normalizes a timestamp value to float seconds.
19         Handles: float/int (pass-through), plain numeric strings ("12.4"),
20         and colon-separated strings ("MM:SS", "HH:MM:SS", "H:MM:SS.ms").
21         This mirrors the parse_time logic used in gemini.py to ensure
22         the stitcher can always consume whatever format the indexer produced."""
23         if isinstance(val, (int, float)):
24             return float(val)
25         if isinstance(val, str):
26             val = val.strip()
27             if ":" in val:
28                 # Handle MM:SS, HH:MM:SS, etc.
29                 parts = val.split(":")
30                 parts.reverse()
31                 total = 0.0
32                 for i, p in enumerate(parts):
33                     try:
34                         total += float(p) * (60 ** i)
35                     except ValueError:
36                         pass
37                 return total
38             try:
39                 return float(val)
40             except ValueError:
41                 logger.warning(f"Could not parse timestamp value '{val}' as a number.
Defaulting to 0.0.")
42                 return 0.0
43                 logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
'{val}'. Defaulting to 0.0.")
44                 return 0.0
45
46     def _get_video_duration(self, video_path: str) -> float:
47         """Probes the video file to get its total duration in seconds.
48         Returns 0.0 if it cannot be determined."""
49         try:
50             result = subprocess.run(
51                 [
52                     "ffprobe", "-v", "error",
53                     "-show_entries", "format=duration",
54                     "-of", "default=noprint_wrappers=1:nokey=1",
55                     video_path
56                 ],
57                 capture_output=True, text=True, check=True
58             )
59             return float(result.stdout.strip())
60         except Exception as e:
61             logger.warning(f"Could not determine video duration via ffprobe: {e}")
62             return 0.0
63
64     def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float,
float]], output_filename: str) -> str:
65         """
66         Extracts selected segments from a raw video and stitches them together
67         into a seamless highlights file, applying end_padding to prevent abrupt endings.
68         """

```

```

69         if not segments:
70             raise ValueError("No highlight segments provided to compile.")
71
72         if not os.path.exists(raw_video_path):
73             raise FileNotFoundError(f"[COMPILER ERROR] Source video file not found:
{raw_video_path}")
74
75         output_path = os.path.join(self.output_dir, output_filename)
76
77         # Probe video duration so we can detect out-of-range segments
78         video_duration = self._get_video_duration(raw_video_path)
79         if video_duration > 0:
80             logger.info(f"Source video duration: {video_duration:.2f}s")
81         else:
82             logger.warning("Video duration unknown. Cannot validate segment boundaries
against video length.")
83
84         # We will create temporary clips and stitch them using FFmpeg concat demuxer
85         temp_clips = []
86         skipped_segments = []
87
88         # Create a temp directory within output_dir
89         with tempfile.TemporaryDirectory(dir=self.output_dir) as tmp_dir:
90             logger.info(f"Trimming {len(segments)} segments
(begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")
91             for idx, (raw_start, raw_end) in enumerate(segments):
92                 # Normalize timestamps to float seconds ? handles float, int,
93                 # plain numeric strings, and colon-separated formats like "MM:SS"
94                 start = self._parse_time(raw_start)
95                 end = self._parse_time(raw_end)
96                 segment_label = f"Segment #{idx+1}/{len(segments)} [{start:.2f}s -
{end:.2f}s]"
97
98                 # Validate the segment times
99                 if start < 0:
100                     logger.warning({segment_label}: start_time is negative
({start:.2f}s). Clamping to 0.)
101                     start = 0.0
102
103                 if start >= end:
104                     logger.error({segment_label}: start_time ({start:.2f}s) >= end_time
({end:.2f}s). Skipping invalid segment.)
105                     skipped_segments.append((idx, start, end, "start >= end"))
106                     continue
107
108                 # Check if segment extends beyond video duration
109                 if video_duration > 0:
110                     if start >= video_duration:
111                         logger.error(f"{segment_label}: start_time ({start:.2f}s) is
beyond the video duration ({video_duration:.2f}s). "
112                                     f"This segment cannot be extracted ? the video ran
out. Skipping.")
113                         skipped_segments.append((idx, start, end, "start beyond video
end"))
114                         continue
115
116                     if end > video_duration:
117                         original_end = end
118                         end = video_duration
119                         logger.warning(f"{segment_label}: end_time ({original_end:.2f}s)
exceeds video duration ({video_duration:.2f}s). "
120                                     f"Clamping end_time to {video_duration:.2f}s. The
segment may be truncated.")
121
122                 clip_path = os.path.join(tmp_dir, f"clip_{idx}.mp4")

```

```

123
124         # Precise sub-second trim using fast re-encoding to preserve stream
headers
125         padded_start = max(0.0, start - self.begin_padding)
126         padded_end = end + self.end_padding
127
128         # Clamp padded_end to video duration if known
129         if video_duration > 0 and padded_end > video_duration:
130             padded_end = video_duration
131             logger.info(f"{segment_label}: Padded end ({end +
self.end_padding:.2f}s) exceeds video duration. "
132                         f"Clamping to {video_duration:.2f}s (end padding
partially applied).")
133
134             trim_duration = padded_end - padded_start
135             logger.info(f"Trimming {segment_label} -> [{padded_start:.3f}s -
{padded_end:.3f}s] (duration: {trim_duration:.2f}s)")
136
137             cmd = [
138                 "ffmpeg", "-y",
139                 "-ss", str(round(padded_start, 3)),
140                 "-to", str(round(padded_end, 3)),
141                 "-i", raw_video_path,
142                 "-c:v", "libx264",
143                 "-preset", "superfast",
144                 "-crf", "22",
145                 "-c:a", "aac",
146                 "-b:a", "128k",
147                 clip_path
148             ]
149
150             try:
151                 result = subprocess.run(cmd, capture_output=True, check=True)
152                 # Verify the output clip was actually created and has content
153                 if os.path.exists(clip_path) and os.path.getsize(clip_path) > 0:
154                     temp_clips.append(clip_path)
155                 else:
156                     logger.error({segment_label}: FFmpeg exited successfully but
output clip is missing or empty. Skipping.)
157                     skipped_segments.append((idx, start, end, "empty output clip"))
158             except subprocess.CalledProcessError as e:
159                 stderr_msg = e.stderr.decode(errors='replace').strip()
160                 logger.error({segment_label}: FFmpeg trim failed.)
161                 print(f"  Command: {' '.join(cmd)}")
162                 print(f"  FFmpeg stderr: {stderr_msg[-500:]}") # Last 500 chars to
avoid flooding
163                 skipped_segments.append((idx, start, end, f"ffmpeg error:
{stderr_msg[-200:]}"))
164                 continue
165
166             # Report summary of skipped segments
167             if skipped_segments:
168                 logger.info(f"\nSUMMARY] {len(skipped_segments)} out of {len(segments)}
segments were SKIPPED:")
169                 for seg_idx, seg_start, seg_end, reason in skipped_segments:
170                     logger.info(f"  - Segment #{seg_idx+1} [{seg_start:.2f}s -
{seg_end:.2f}s]: {reason}")
171                 logger.info("")
172
173             if not temp_clips:
174                 raise RuntimeError(
175                     f"[COMPILER ERROR] Failed to generate ANY valid highlight clips. "
176                     f"All {len(segments)} segments were skipped. "
177                     f"Check the errors above for details. "
178                     f"Common causes: video file is corrupted, segments are out of range,

```

```

or ffmpeg is not installed."
179         )
180
181         logger.info(Successfully trimmed {len(temp_clips)}/{len(segments)} clips.
Preparing to stitch...)
182
183         # Create a list file for FFmpeg concat demuxer
184         list_file_path = os.path.join(tmp_dir, "clips.txt")
185         with open(list_file_path, "w") as f:
186             for clip in temp_clips:
187                 # Escape single quotes in path
188                 safe_path = clip.replace("'", "'\\'")
189                 f.write(f"file '{safe_path}'\n")
190
191         logger.info(Stitching clips together...)
192         # Stitch them together without re-encoding (since they were encoded with
identical settings)
193         concat_cmd = [
194             "ffmpeg", "-y",
195             "-f", "concat",
196             "-safe", "0",
197             "-i", list_file_path,
198             "-c", "copy",
199             output_path
200         ]
201
202         try:
203             result = subprocess.run(concat_cmd, capture_output=True, check=True)
204
205             # Final verification
206             if not os.path.exists(output_path) or os.path.getsize(output_path) == 0:
207                 raise RuntimeError(
208                     f"[COMPILER ERROR] FFmpeg concatenation appeared to succeed, "
209                     f"but output file is missing or empty: {output_path}"
210                 )
211
212             output_size_mb = os.path.getsize(output_path) / (1024 * 1024)
213             logger.info(Dynamic highlight reel saved: {output_path}
({output_size_mb:.1f} MB))
214             return output_path
215         except subprocess.CalledProcessError as e:
216             stderr_msg = e.stderr.decode(errors='replace').strip()
217             raise RuntimeError(
218                 f"[COMPILER ERROR] FFmpeg concatenation failed.\n"
219                 f"Command: {' '.join(concat_cmd)}\n"
220                 f"FFmpeg stderr: {stderr_msg[-500:]}"
221             )
222

```

5. user (2026-06-10T05:50:10.872Z)

Switched to a new branch 'fix/compiler-fstring-syntax'

```

        print(f"[COMPILER WARNING] Could not parse timestamp value '{val}' as a
number. Defaulting to 0.0.")
        print(f"[COMPILER WARNING] Unexpected
timestamp type {type(val).__name__} for
value '{val}'. Defaulting to
0.0.")
        print(f"[COMPILER WARNING] Could not determine video duration via ffprobe:
{e}")
        print(f"[COMPILER] Source video duration: {video_duration:.2f}s")
        print("[COMPILER WARNING] Video duration
unknown. Cannot validate segment
boundaries against video

```

```

length.")
    print(f"[COMPILER] Trimming {len(segments)}
segments (begin_padding={self.begin_padding}s,
end_padding={self.end_padding}s)...")
    print(f"[COMPILER WARNING] {segment_label}: start_time is negative
({start:.2f}s). Clamping to 0.")
    print(f"[COMPILER ERROR] {segment_label}:
start_time ({start:.2f}s) >= end_time
({end:.2f}s).
Skipping invalid segment.")
    print(f"[COMPILER ERROR]
{segment_label}: start_time ({start:.2f}s) is
beyond the video
duration ({video_duration:.2f}s). "
    print(f"[COMPILER WARNING]
{segment_label}: end_time ({original_end:.2f}s)
exceeds video
duration ({video_duration:.2f}s). "
    print(f"[COMPILER INFO] {segment_label}:
Padded end ({end +
self.end_padding:.2f}s) exceeds video
duration. "
    print(f"[COMPILER] Trimming {segment_label}
-> [{padded_start:.3f}s - {padded_end:.3f}s]
(duration:
{trim_duration:.2f}s)")
    print(f"[COMPILER ERROR]
{segment_label}: FFmpeg exited successfully
but output clip is
missing or empty. Skipping.")
    print(f"[COMPILER ERROR] {segment_label}: FFmpeg trim failed.")
    print(f"  Command: {' '.join(cmd)}")
    print(f"  FFmpeg stderr: {stderr_msg[-500:]}") # Last 500 chars to
avoid flooding
    print(f"\n[COMPILER SUMMARY] {len(skipped_segments)} out of
{len(segments)} segments were SKIPPED:")
    print(f"  - Segment #{seg_idx+1} [{seg_start:.2f}s -
{seg_end:.2f}s]: {reason}")
    print()
    print(f"[COMPILER] Successfully trimmed {len(temp_clips)}/{len(segments)}
clips. Preparing to stitch...")

```

6. user (2026-06-10T05:50:26.180Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

7. user (2026-06-10T05:50:28.050Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

8. user (2026-06-10T05:50:30.253Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

9. user (2026-06-10T05:50:31.345Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is

current in your context ? no need to Read it back)

10. user (2026-06-10T05:50:32.782Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

11. user (2026-06-10T05:50:34.641Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

12. user (2026-06-10T05:50:36.053Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

13. user (2026-06-10T05:50:37.233Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

14. user (2026-06-10T05:50:52.215Z)

backend.main import OK

15. user (2026-06-10T05:50:58.151Z)

```
1      #!/usr/bin/env python
2      """Run the FULL test suite for the badminton-highlight-indexer.
3
4      A single, discoverable entry point so you (or a reviewer / CI) can verify the
5      whole suite before opening a PR:
6
7          python run_all_tests.py          # full suite, quiet
8          python run_all_tests.py -v       # verbose (passthrough to pytest)
9          python run_all_tests.py -k config # filter by keyword (passthrough)
10         python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12     Anything after the script name is passed straight through to pytest.
13
14     Notes
15     -----
16     - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17     - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18       package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19         pip install -e ../sports-obsreport --no-build-isolation
20     - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21     """
22     from __future__ import annotations
23
24     import importlib.util
25     import subprocess
26     import sys
27
28
29     def _installed(module: str) -> bool:
30         try:
31             return importlib.util.find_spec(module) is not None
32         except (ImportError, ValueError):
```

```

33         return False
34
35
36     def main() -> int:
37         args = list(sys.argv[1:])
38         # Default to quiet unless the caller chose a verbosity.
39         if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40             args = ["-q", *args]
41         cmd = [sys.executable, "-m", "pytest", *args]
42
43         obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44         bar = "=" * 72
45         print(bar)
46         print(" Full test suite: " + " ".join(cmd))
47         print(f" obsreport (optional reporting dep): {obs}")
48         print(bar, flush=True)
49
50         rc = subprocess.run(cmd).returncode
51
52         print(bar)
53         print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
54         print(bar)
55         return rc
56
57
58     if __name__ == "__main__":
59         raise SystemExit(main())
60

```

16. user (2026-06-10T05:50:58.383Z)

```

tests\__init__.py
tests\test_eval_annotations.py
tests\test_eval_batch.py
tests\test_eval_classifier.py
tests\test_eval_metrics.py
tests\test_eval_training.py
tests\test_geometry.py
tests\test_providers.py
tests\test_sport_profiles.py
tests\test_video.py
tests\test_annotation_providers.py
tests\test_calibration.py
tests\test_calibrate_wasb.py
tests\test_export_wasb_segments.py
tests\test_gemini_refine.py
tests\test_provider_gemini.py
tests\test_calibrate_local.py
tests\test_distill_local.py
tests\test_automated_provider.py
tests\test_database.py
tests\test_eval_harness.py
tests\test_eval_regression.py
tests\test_gemini.py
tests\test_hit_detector.py
tests\test_rally_gate.py
tests\test_reporting_harvest.py
tests\test_tracknet_hybrid.py
tests\test_validators_base.py
tests\test_wasb_cache.py
tests\test_yolo_hybrid.py
tests\test_base.py
tests\test_config.py

```



```
tests\test_hashing.py
tests\test_rally_slicer.py
tests\test_reporting.py
tests\test_reporting_spec_faq.py
tests\test_tracknet_runner.py
tests\test_eval_partial_labels.py
tests\test_rally_seq_proto.py
```

17. user (2026-06-10T05:51:14.884Z)

```
alias(name='os')
alias(name='sys')
alias(name='argparse')
alias(name='json')
alias(name='logging')
alias(name='uvicorn')
from dotenv import ...
from fastapi import ...
from fastapi.middleware.cors import ...
from pydantic import ...
from typing import ...
from fastapi.staticfiles import ...
from fastapi.responses import ...
from backend.pipeline.validators.base import ...
from backend.infrastructure.database import ...
from backend.pipeline.segmenters.base import ...
from backend.pipeline.stitcher.compiler import ...
from backend.utils.hashing import ...
from backend.config import ...
from backend.results import ...
from backend.reporting import ...
alias(name='cv2')
alias(name='subprocess')
```

18. user (2026-06-10T05:51:15.025Z)

```
Found 14 files
backend\eval\rally_seq_proto.py
backend\pipeline\segmenters\wasb_hybrid.py
backend\main.py
backend\pipeline\detectors\wasb_infer.py
backend\pipeline\detectors\wasb_runner.py
backend\pipeline\segmenters\motion.py
backend\pipeline\segmenters\tracknet_hybrid.py
backend\pipeline\validators\scene_cut.py
backend\pipeline\validators\relevance.py
backend\pipeline\validators\blur.py
backend\pipeline\segmenters\yolo_hybrid.py
backend\pipeline\audio\hit_detector.py
backend\eval\distill_local.py
backend\eval\classifier.py
```

19. user (2026-06-10T05:51:43.347Z)

```
1      """Tests for the canonical video_id hashing helper.
2
3      Guards the de-duplication of the former 4 divergent get_file_md5 copies
4      (backend/main.py + the three run_*.py scripts) into one shared function.
5      """
6
7      import hashlib
8      import os
9      import tempfile
10
```

```

11     from backend.utils.hashing import compute_video_id
12
13
14     def test_compute_video_id_matches_plain_md5():
15         """The id is exactly the MD5 of the whole file (chunk size must not affect it)."""
16         data = b"the quick brown fox jumps over the lazy dog" * 1000
17         with tempfile.TemporaryDirectory() as td:
18             p = os.path.join(td, "clip.mp4")
19             with open(p, "wb") as f:
20                 f.write(data)
21             assert compute_video_id(p) == hashlib.md5(data).hexdigest()
22
23
24     def test_compute_video_id_is_stable():
25         with tempfile.TemporaryDirectory() as td:
26             p = os.path.join(td, "clip.mp4")
27             with open(p, "wb") as f:
28                 f.write(b"\x00\x01\x02" * 5000)
29             assert compute_video_id(p) == compute_video_id(p)
30
31
32     def test_compute_video_id_changes_with_content():
33         with tempfile.TemporaryDirectory() as td:
34             a = os.path.join(td, "a.mp4")
35             b = os.path.join(td, "b.mp4")
36             with open(a, "wb") as f:
37                 f.write(b"aaaa")
38             with open(b, "wb") as f:
39                 f.write(b"aaab")
40             assert compute_video_id(a) != compute_video_id(b)

```

20. assistant (2026-06-10T05:51:56.775Z)

"""Import/syntax smoke tests for the backend package.

Guards against a SyntaxError (or module-level import error) shipping while the suite stays green: most of the suite imports individual backend modules, so a broken file that nothing imports (e.g. the stitcher compiler, which only backend.main pulls in) used to go unnoticed.

Both tests are offline and dependency-light: the syntax sweep never imports anything, and the backend.main import runs in a subprocess with any missing heavy deps (cv2/torch/numpy) stubbed, so they also pass in environments without the GPU stack.

"""

```

import py_compile
import subprocess
import sys
from pathlib import Path

```

```

REPO_ROOT = Path(__file__).resolve().parents[1]

```

```

def test_backend_package_has_no_syntax_errors():
    """Every .py file under backend/ must byte-compile."""
    failures = []
    for path in sorted((REPO_ROOT / "backend").rglob("*.py")):
        try:
            py_compile.compile(str(path), doraise=True)
        except py_compile.PyCompileError as exc:
            failures.append(f"{path.relative_to(REPO_ROOT)}: {exc.msg}")
    assert not failures, "Syntax errors in backend/:\n" + "\n".join(failures)

```

```
def test_backend_main_imports():
    """backend.main (FastAPI app, /api/compile) must be importable.

    Runs in a subprocess so stubbing missing heavy deps cannot leak into this
    pytest process; cwd=repo root puts the package on sys.path.
    """
    prelude = (
        "import importlib.util, sys\n"
        "from unittest import mock\n"
        "for name in ('cv2', 'torch', 'numpy'):\n"
        "    if importlib.util.find_spec(name) is None:\n"
        "        sys.modules[name] = mock.MagicMock()\n"
        "import backend.main\n"
    )
    result = subprocess.run(
        [sys.executable, "-c", prelude],
        cwd=str(REPO_ROOT),
        capture_output=True,
        text=True,
    )
    assert result.returncode == 0, (
        "import backend.main failed:\n" + result.stderr.strip()
    )
)
```

21. user (2026-06-10T05:51:56.795Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py (file state is current in your context ? no need to Read it back)

22. assistant (2026-06-10T05:52:01.317Z)

Now running the new tests, then the full suite:

23. user (2026-06-10T05:52:07.693Z)

```
===== test session starts =====
platform win32 -- Python 3.13.13, pytest-9.0.3, pluggy-1.6.0 --
C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\avidu\Projects\badminton-highlight-indexer
configfile: pytest.ini
plugins: anyio-4.13.0, cov-7.1.0
collecting ... collected 2 items

tests/test_import_smoke.py::test_backend_package_has_no_syntax_errors PASSED [ 50%]
tests/test_import_smoke.py::test_backend_main_imports PASSED [100%]

===== 2 passed in 4.19s =====
```

24. assistant (2026-06-10T05:52:15.100Z)

Both pass. Quick negative check that the guard would have caught the original bug, then the full suite:

25. user (2026-06-10T05:52:17.117Z)

```
Exit code 1
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import sys; compile(sys.stdin.read(), 'compiler.py', 'exec')
    ~~~~~^~~~~~
File "compiler.py", line 100
    logger.warning({segment_label}: start_time is negative ({start:.2f}s). Clamping to 0.)
```

SyntaxError: invalid decimal literal

26. user (2026-06-10T05:52:40.448Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 28%]
..... [ 56%]
..... [ 84%]
..... [100%]
256 passed in 15.70s
=====
RESULT: PASS - all tests green
=====
```

27. assistant (2026-06-10T05:52:46.075Z)

All 256 tests pass (254 existing + 2 new). Committing and opening the PR:

28. user (2026-06-10T05:52:55.184Z)

```
warning: in the working copy of 'tests/test_import_smoke.py', LF will be replaced by CRLF the
next time Git touches it
[fix/compiler-fstring-syntax e14df4e] fix(stitcher): restore f-string quotes dropped in logging
conversion
 2 files changed, 63 insertions(+), 8 deletions(-)
 create mode 100644 tests/test_import_smoke.py
remote:
remote: Create a pull request for 'fix/compiler-fstring-syntax' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/compiler-
fstring-syntax
remote:
branch 'fix/compiler-fstring-syntax' set up to track 'origin/fix/compiler-fstring-syntax'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/compiler-fstring-syntax -> fix/compiler-fstring-syntax
```

29. user (2026-06-10T05:53:12.260Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/63>

30. user (2026-06-10T05:53:20.151Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-475 of 1069 total (47726 tokens, cap 25000). Call Read with offset=476 limit=475 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit
```

```

`wf_2a64bc88-2a8` over indexer+collector:
13     engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors
completed+verified, 5 rate-limited).
14     - ? **CRITICAL ? master is BROKEN since 7fbbbe0 (2026-06-07 print?logger sweep):**
15     `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines
(100/104/156/160/181/191/213)** ? SyntaxError ?
16     `import backend.main` fails ? FastAPI server + `/api/compile` dead. Triple-verified +
reproduced first-hand
17     (`py_compile` exit 1). The **254-green suite never noticed** (zero tests import
backend.main / compiler / wasb_hybrid /
18     motion / any concrete validator; **no CI exists at all**). Spawned fix chip
`task_2b7a19d4` (fix + import/compileall
19     smoke test). Coverage is INVERTED vs production risk (tracknet twin 8 tests vs live
wasb_hybrid 0; TrackNetRunner 21
20     vs WasbRunner 1). Suite itself healthy where it exists (254 green / 19.9s / faithful
mocks / low flake risk); "fully
21     mocked runs anywhere" is false on cv2-free machines (bare `import cv2` at collection
time, only obsreport importorskips).
22     - **Doc drift (verified, one-directional ? newest decisions not back-propagated):** (1)
licensing "Apache-2.0 only" still
23     in CLAUDE.md:33 / PLATFORM:269,301 / COMMERCIALIZATION C14 vs ratified
**MIT/Apache/BSD** (a literal CLAUDE.md reader
24     rejects the plan's own MIT ActionFormer/ASFormer heads); (2) **Gemma-4 still
"approved" in those same docs vs
25     AMBER/bench-only in the plan; (3) **M0.3 purge pointer WRONG** ?
`training.label_candidates` is a pure GT-agnostic fn;
26     the live Gemini-silver training violation = `backend/eval/distill_local.py` (+
calibrate_local thresholds) + gitignored
27     `output/local_rally_model.json`; (4) **C3 escape hatch broken** ? "train-own weights
(Phase-4 roadmap)" cites a phase
28     that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T
train a detector) ? own-weights is
29     an unscoped new workstream (coordinate labels or detector swap); (5)
OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are
30     linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS
is branch-only; (6) README
31     "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default";
(7) ROADMAP still says audio
32     "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve";
(8) Roora multisport CSVs live ONLY
33     in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline
TT-0.909/pickleball-0.308 evidence is one cleanup from
34     loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per
sport"); (10) ship-gate F1@tIoU still
35     undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no
tracker (BACKLOG lacks it).
36     - **PMF (web-grounded, sources in audit output):** pricing/cost benchmarks VERIFIED
(SwingVision $179.99/yr ~20k paying
37     subs rev $2.75M +12% YoY; PB Vision $99/yr + **$8/hr-of-video partner API**; Gemini
COGS ~$0.30?0.60/hr ? fat margin);
38     but **"India badminton whitespace" FALSIFIED ? Bangalore is the most contested
badminton-AI city** : Machaxi ($1.5M
39     Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School),
Game Theory (Gopichand-backed smart
40     courts w/ auto-highlights), Stupa ($3.3M, BWF-approved, TT?badminton) ? NONE in our
2026-06-07 competitive table.
41     Surviving wedge (narrow, real): **software-only rally-precise indexing of EXISTING
messy multi-court BYO footage**
42     (market solves it only with per-court hardware) sold per-hour as engine/API, possibly
THROUGH those players
43     (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) +
**Gemini-wrapper baseline vs the
44     n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid
academy pilot.
45     - **5 auditors PENDING (rate limit, resets ~15:50 IST): eng-pipeline, eng-eval,

```

```

reliability, collector, docs-technical.**
46     Resume (cached prefix): `Workflow({scriptPath: "C:\\Users\\avidu\\.claude\\projects\\C
--Users-avidu-Projects-sports-data-
collector\\45ea3a91-654d-4972-9cd3-65f35db54caf\\workflows\\scripts\\deep-project-audit-
wf_2a64bc88-2a8.js", resumeFromRunId: "wf_2a64bc88-2a8"})`.
47     Full audit JSON: `%LOCALAPPDATA%\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\tasks\wf38bovts.output`.
48
49     **Checkpoint:** 2026-06-09 (Roora MULTISPORT recon ? for FRESH next session) . ??
**Pickleball + Tabletennis golden labels
50     arrived; INGESTION = SOLVED, but the "merged model" experiment is BLOCKED on per-sport
trajectory detectors (validates the
51     roadmap blocker).** Owner left a recon-only ask (deliberately did NOT drain context /
spin heavy jobs ? owner wants me fresh).
52     - **Deliverables:** `Annotation Setup/6 June 2026/{pickleball,tabletennis}/*.zip` (3
CVAT export FORMATS each: COCO
53     `instances_default.json`, **CVAT-XML `annotations.xml`** ? use this, Pascal-VOC).
Videos: `~/6 June 2026/Annotation Videos/
54     Video 2 - Pickleball.mp4` (4K, 59.94fps, 97s, 5737fr) + `Video 3 - Tabletennis.mp4`
(4K, 59.94fps, 58s, 3490fr). Short clips.
55     - **What Roora annotated:** label class "rally" as **per-frame BOUNDING BOXES on sampled
frames (~every 30fr)** ? box geometry
56     is just a CARRIER; the boxes carry the FULL rally schema as CVAT **attributes**
(rally_number, start/end time, shot_count,
57     score_before/after(A-B), ending_reason, last_shot_outcome [+note for TT]). 456/513
attr instances. Boxed frames cluster into
58     rallies (runs of ~30-spaced frames = a rally; big gaps = dead time).
59     - **INGESTION = YES, cleanly:** the collector's `src/parsers/cvat/rally_cvat.py`
(`curate convert-cvat`) is PURPOSE-BUILT for
60     exactly this ? ignores box geometry, **recomputes rally interval from the frame
range** (`sec=frame/fps` at true fps, ignoring
61     vendor start/end times which are unreliable), groups boxes by rally_number, maps
attrs?canonical. Minor: verify the attr-name
62     mapping handles spaces/"(A-B)" (`ATTR_MAP`). Collector already has pickleball +
tabletennis schemas.
63     - ? **"MERGED LEARNT MODEL" across sports is BLOCKED:** `rally_seq_proto` features come
from a WASB **shuttle** trajectory =
64     BADMINTON-ONLY; pickleball (wiffle ball) + TT (tiny fast ball on a table) are out-of-
distribution ? no clean trajectory yet.
65     This EMPIRICALLY validates the roadmap's binding multisport blocker (clean per-sport
MIT/Apache ball detectors not identified).
66     - **PROGRESS (owner asked to complete 1/2/3 before badminton):**
67     - ? **(1) convert-cvat DONE ? INGESTION PROVEN.** `python -m src.cli.curate convert-
cvat --input <annotations.xml> --out
68     <csv> --fps 59.94` ? **pickleball 6 rallies + tabletennis 6 rallies**, sane
intervals, attrs mapped. CSVs at
69     `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv` (+ extracted XMLs
`%TEMP%\roora\{pb,tt}\annotations.xml`). ? Roora data
70     nits (non-blocking): pickleball `last_shot_outcome='0'` (off-vocab in/n_a/net/out);
TT `ending_reason` mostly 'other' with the
71     real outcome in `notes` ("out of table"/"net") ? ending_reason taxonomy mapping
could improve. Both clips SHORT (97s/58s).
72     - ? **(2) WASB-transfer test DONE (`%TEMP%\{pickleball,tabletennis}_traj.csv`; log
`~/clips/multisport_run.log`):** WASB
73     (badminton shuttle detector) detects SOMETHING on both ? **pickleball 28% /
tabletennis 39% visibility** (NOT near-zero;
74     comparable to badminton 40?67%). So it's not blind.
75     - ? **(3) DONE ? SURPRISE RESULT (directional, n=6 each, TINY so noisy):**
`eval_partial_labels` on (WASB-transfer traj + Roora
76     6-rally CSV, fps 59.94, fw 1920): **TABLE TENNIS best F1 0.909** (tuned+gate2:
P1.0/R0.833, 5/6 matched, boundary err 0.49s ?
77     **WASB transfers to TT remarkably well!**); **PICKLEBALL best F1 0.308** (poor ?
wiffle ball too out-of-distribution).
78     ? **NUANCES the roadmap blocker: per-sport detector transfer is SPORT-DEPENDENT ? TT
may be servable with WASB short-term;

```

79 pickleball needs a proper detector.** ? BIG caveat: 6 rallies each = could be partly
 luck ? owner sending more videos to

80 confirm. A "merged" prototype LOVO across badminton+pickleball+TT is the natural
 next experiment once n grows per sport.

81 Roor LABELS ingest cleanly + are corpus-gold regardless. **Great honest surprise to
 greet owner with.**

82

83 **Checkpoint:** 2026-06-09 (PR #61 MERGED ? roadmap SET, session wrap) . ? **Owner
 agreed + merged #61 (`master 4d04a73`).

84 ALL CLEAN: master, 0 uncommitted, 0 open PRs, branches pruned.** The owned-model
 ROADMAP is now set + agreed ?

85 authoritative doc = **`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`** (full verified
 decisions in the "(PR review)" checkpoint

86 just below: ms-swift + Qwen3-VL, TAL-head-first, conversational-QA seam-now-model-later,
 multisport one-head/base, Python

87 everywhere, MIT/Apache/BSD ratified, "envision now build later"). [[session-handoff]]
 rewritten to this phase.

88 - ?? **NEXT SESSION (awaiting owner greenlight ? NO build until then):** greenlight a
 Phase-0 milestone (rec **M0.1

89 corpus+event-index** ? **M0.4 Gen-0 TAL harness**, both \$0/local); confirm **collector
 = system-of-record** ; define the

90 **ship-gate F1@tIoU target** (floor = beat heuristic LOVO 0.480); **rename repo off
 "badminton"**, M0.3 purge Gemini

91 training labels + WASB C3 (own-weights/clean swap). Corpus = n=6
 (`output/human\_lovo\_manifest.json`); learned prototype

92 is the Gen-0 seed. Committed next PLATFORM slice stays the async job model (single-
 tenant). Owner flagged context ~85% ? this is the wrap checkpoint.

93

94 **Checkpoint:** 2026-06-09 (PR review) . ? **OWNER REVIEWED PR #61 ? REFRAMED ROADMAP
 (important checkpoint).** Owner left

95 ~12 inline comments (mostly state-updates + recommendations wanted). **Key roadmap
 shifts to internalize:**

96 - **Immediate goal REFRAMED:** not "on-device model" but a **TRAINING FRAMEWORK +
 HARNESS to produce a VERY HIGH

97 precision/recall model that extracts HIGHLIGHTS + RALLIES from a clip.** Quality
 first. On-device = LATER (compression

98 for power users to save upload cost), definitely on roadmap but not now.

99 - **MULTISPORT FROM THE START** ? all net-separated racquet sports ? eventually all
 sports; a SINGLE sports-agnostic model.

100 Collector + indexer already sport-generic by design. ? **ACTION: rename the repo off
 "badminton" ASAP** (owner emphatic).

101 - **CONVERSATIONAL VIDEO-QA = north-star feature:** upload video ? ask contextual
 questions ("longest rally", "make a clip

102 of all service faults", "rally >20 shots"). Needs per-video STATE/bookkeeping ? design
 the framework for it. Owner asks:

103 embark now or punt + revisit after quality? (My lean: design the bookkeeping/event-
 store layer now, punt the heavy

104 conversational model ? pending the strategy workflow.)

105 - **LICENSING GUARDRAIL (owner: "never corrupted")** EVERY dependency ? code, data, AND
 weights ? must be usable in

106 PRIVATE PROPRIETARY COMMERCIAL software. **MIT / Apache-2.0 / BSD only.** Ratify (was
 "Apache-2.0 only"). Enforce religiously.

107 - Owner wants a **coherent recommendation: local Gemma 4 vs ms-swift starting point**
 (multisport, single, conversational,

108 commercial). NB: ms-swift = training FRAMEWORK, Gemma 4 = BASE model ? orthogonal;
 clarify in the reply.

109 - Owner: **upgrade WASB C3 (own-weights/provenance) to a first-class action item now.**
 M0.2 (grow n) = practically DONE

110 (n=6); owner will add 1 more varied multi-court-club video soon ? Phase-0 ready to
 start.

111 - **LANGUAGE QUESTION (owner callout, no preference):** is Python right for the upcoming
 work, or brainstorm alternatives?

112 - ? **STRATEGY WORKFLOW done (`wf\_f9aa1d1c-c83`, 10 agents, verified+critiqued) ?
 VERIFIED DECISIONS (now in the rewritten

113 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, the authoritative roadmap; pushed to PR #61

```

+ a consolidated PR comment reply):**
114     - **Framework = ms-swift (Apache-2.0)** (only forge w/ native video+audio+timestamp-
grounding+video-GRP0); unsloth = 8GB-local
115     SFT/compression tool (avoid its AGPL Studio). **Base = Qwen3-VL (Apache-2.0,
timestamp emission)** primary; **Gemma-4 STAYS
116     AMBER/bench-only** (critic corrected the workflow's "clear AMBER" ? posture
unconfirmed + 60s/no-timestamp dispute);
117     InternVL3 (MIT) fallback. ms-swift?Gemma4 (framework vs base ? clarified for owner).
118     - **START WITH THE TAL HEAD, NOT THE VLM** (VLMs miss strict temporal boundaries;
boundary accuracy = the product; head
119     de-risked AUC 0.83?0.92, $0 local, doubles as teacher+baseline). Head now ? VLM
later (two-stage: VLM proposes, head refines).
120     - **Conversational-QA: build the SEAM now (per-video event store + structured text-to-
SQL + ffmpeg clip-cut), PUNT the NL
121     model** until quality clears the bar. Queries ("longest rally"/"service faults"/">20
shots") = structured-numeric ? SQL, not
122     a VLM. Reserve fault/event taxonomy in M0.1. ? real DB delta (new cols + a
`highlights` table; play_segments lacks them today).
123     - **Multisport: one shared head + one base; scale DATA + per-sport clean trajectory
detectors (the binding blocker ? none
124     identified for tennis/TT/pickleball/padel yet); WASB C3 multiplies. Single-model
temporal generalization = OPEN HYPOTHESIS
125     (RacketVision evidence REFUTED = ball-tracking not temporal). Repo rename off
"badminton" = action item.**
126     - **Language: Python everywhere** (training Python-locked; platform I/O-bound, no hot
path ? reject Go/Rust split); on-device =
127     deferred EXPORTED runtime shell (ONNX/CoreML/GGUF), doesn't dictate dev language.
Constraint NOW: **train export-clean ops**.
128     - **Honest gaps:** cost-to-Gen-2 UNQUANTIFIED (biggest risk); ship-gate "very high
P/R" target undefined (floor = beat heuristic
129     LOVO 0.480 ? set explicit F1@tIoU); per-sport detectors TBD; base PRETRAINING
provenance unauditable ? owner risk-acceptance;
130     **Gemini-silver TRAINING labels = LIVE violation ? hard-blocker purge M0.3.**
131     - **Owner principle recorded: "envision now, build later"** ? design every deferred
seam (event store, export boundary, sport-
132     agnostic head) up front; punt the feature, never the seam.
133     - ?? **NEXT:** owner reviews the reframed plan + decisions (greenlight a Phase-0
milestone ? rec M0.1 corpus+event-index then M0.4
134     harness; confirm collector=system-of-record). Then start the greenlit milestone. Repo
rename. Define the ship-gate F1@tIoU target.
135
136     **Checkpoint:** 2026-06-09 . ? **n=6: MahadevpuraSingles added while owner reviews PR
#61.** Owner moved ALL golden videos
137     + CSVs to `Annotation Setup/Collect/Golden Labelled/` (manifest label paths updated;
trajectories unaffected in `%TEMP%`).
138     New: **MahadevpuraSingles** (singles, 1080p, 29.97fps, 31 rallies; traj
`%TEMP%mahadevpura_traj.csv`, script
139     `run_wasb_mahadevpura.sh`). **Borderline multi-court (contrast 2.2x)** ? heuristic best
0.667, learned held-out **AUC
140     0.902/F1 0.712** (clean generalization to an unseen match). **n=6 LOVO: heuristic 0.480
± 0.242 vs learned 0.583** (learned
141     holds ~+0.10 since n=3). Contrast metric now predicts heuristic F1 MONOTONICALLY across
5 matches (Mahadevpura's 2.2x?0.67
142     lands on the curve). **Decoupling headline: learned held-out AUC 0.83?0.92 regardless of
contrast; heuristic swings 0.16?0.75
143     with it.** Committed to PR #61 (n=6 doc update). ? GOTCHA: don't combine inner `&` with
run_in_background (orphans the job);
144     calibrate_local buffers output to end + is slow (~17?20min). "Ingested" = added to EVAL
corpus (manifest); collector
145     import-local (system-of-record) still = M0.1, plan-only pending greenlight. ?? owner
reviewing PR #61 (4 plan decisions pending).
146
147     **Checkpoint:** 2026-06-08 (later+4, SESSION WRAP) . ? **n=5 corpus + MULTI-COURT root
cause + heuristic discovery + SINGLE PR #61 for offline review.**
148     Owner tagged gBaaddy (47, 1080p, multi-court) + TestLargeVideo (6, short) ? corpus

```



```

**n=5** (`output/human_lovo_manifest.json`).
149 - **HEURISTIC DISCOVERY (user asked "discover more heuristics like audio"):** tested 2
label-free trajectory pre-filters
150 for multi-court (measured on all 5): foreground spatial-density gate = **net 0.04**
(helps multi-court +0.03, hurts clean
151 0.12); track-continuity/jump filter = **no-op**. ? simple single-cue trajectory
heuristics CAN'T isolate the foreground
152 court. **AUDIO-foreground idea PROTOTYPED (no more videos needed) ? WEAK ?:** on the 2
multi-court videos + clean
153 control, audio onset/RMS energy AUC only **0.53?0.65** for in-rally vs dead-time; HF
bands (8?22kHz @48kHz) raise
154 CONTRAST (Kushagra 1.1?1.7x, gBaaddy 1.3?1.9x, confirming distance-attenuation
physics) but AUC stays ~0.6. Fails:
155 dead-time isn't quiet ? foreground court loud during dead-time too
(footwork/talk/pickup) + gym reverb ? audio loudness
156 ? court occupancy, not rally (same wall as E1 ~2.2x). **Audio alone ? multi-court
gate.** Remaining heuristic lever =
157 real court-region detection (player-cluster/court-line, ROADMAP #2); robust path = the
learned visual model (AUC 0.85).
158 All tested + corrected in PR #61 doc + a PR comment.
159 - **COURT-REGION DETECTION SMOKE-TESTED ? LOW HEADROOM ?:** only 12?14% of dead-time
detections on the multi-court videos
160 fall OUTSIDE the foreground rally region (86?88% INSIDE) ? a perfect court gate lifts
contrast only ~1.6?2.2x (still
161 failing). The confusion is the foreground court's OWN between-point activity, NOT
separable background courts. So court
162 detection (heuristic OR own model) won't help. **3 heuristics now tested + FAILED
(spatial gate, audio, court
163 separability) ? all because it's a TEMPORAL rally-vs-non-rally problem in the same
space; only the learned model solves
164 it (AUC 0.83). NO cheap heuristic shortcut ? the learned segmenter is the path.** (In
PR #61 doc + PR comment.)
165 - ? **SINGLE PR #61** (https://github.com/avidullu/badminton-highlight-indexer/pull/61)
carries EVERYTHING for offline
166 review: rally_seq_proto + tests, owned-model STUDY + PLAN (4 decisions pending),
multivideo-lovo learnings (n=5 + multi-court
167 + heuristic discovery), CODE_MAP. Title/desc updated. Plan doc NOW committed (was
held). Temp experiment scripts deleted.
168 (#60 + collector #13 merged earlier; #61 = only open PR.)
169 - **? DEFINING FINDING (owner's insight, data-confirmed): heuristic fails on MULTI-COURT
footage.** gBaaddy (1080p like
170 Adarsh) heuristic best F1 **0.277**; Kushagra 0.163 ? both have BACKGROUND GAMES. WASB
tracks every shuttle incl. other
171 courts ? high dead-time visibility (gBaaddy 41%, Kushagra 57% vs clean Adarsh 18% /
doubles 22%). **"Contrast" metric**
172 (in-rally=dead-time vis) predicts heuristic success: ?3.6x works, ?1.9x fails. Academy
= multi-court = TARGET ENV ? these
173 are the most representative videos. **Lever = foreground-court isolation** (spatial
gate to prominent court; needs real
174 court-region detection, naive centroid muddled by foreground-shuttle-resting-low).
NEAR-TERM, no new model.
175 - **Learned prototype LOVO (n=5):** per-frame AUC 0.83?0.92 on every match; **recovers
BOTH multi-court videos**
176 (Kushagra 0.561, gBaaddy 0.574 vs heuristic 0.16/0.28). Kushagra climbs
0.049?0.417?0.561 with n. Mean 0.568 ? dipped
177 from 0.615 (n=3) ONLY due to the 6-rally TestLarge noise fold; **excl. it = 0.639**.
Per-video gap on hard videos is the signal.
178 - **Heuristic LOVO n=5 = 0.463 ± 0.247** (per-fold: Adarsh 0.751, doubles 0.758,
Kushagra 0.157, gBaaddy 0.295, testlarge
179 0.353). **FELL from n=3's 0.544** as multi-court videos entered ? the heuristic can't
represent the target env, so a more
180 representative corpus LOWERS its honest score. vs **learned 0.568** ? gap INVERTED
(heuristic led at n=2 +0.066; learned
181 leads at n=5 **+0.105** and widening). (b8jpxbryo was just slow, ~17min; completed
fine.)

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```

182 - **SESSION WRAP shipped ? PR #61** (branch feat/owned-model-study-and-prototype): n=5 +
multi-court finding appended to
183 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`; **session-handoff.md
REWRITTEN** (quality-iteration phase);
184 **CODE_MAP updated** (eval_partial_labels + rally_seq_proto + multi-court gotcha).
Owned model still PLAN-ONLY
185 (`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` DRAFT, 4 decisions pending).
186 - ?? **NEXT (tomorrow): owner tags more varied videos** ? re-run LOVO+prototype.
**Foreground-court isolation experiment**
187 = #1 lever. Owner reviews plan + greenlights a Phase-0 milestone (rec M0.1 corpus
spine). Commit plan doc on approval.
188
189 **Checkpoint:** 2026-06-08 (later+3) . ? **n=3 CROSSOVER ? learned model OVERTAKES the
heuristic (the data-backed green
190 light).** User tagged a 3rd distinct match **LargeTestVideo (DOUBLES, 5.3K?1920,
29.97fps, 49 rallies)** ? corpus now
191 n=3 (2 singles + 1 doubles, 3 cameras/frame-rates). Trajectory
`~/clips/largetestvideo_traj.csv` (script
192 `run_wasb_largetestvideo.sh`). Manifest `output/human_lovo_manifest.json` now n=3.
193 - **Single-video LargeTest (doubles):** heuristic best F1 **0.750** (tuned, R0.918),
best-fit 0.827 ? **doubles is
194 heuristic-FRIENDLY; Kushagra remains the LONE heuristic failure** (not a play-type
issue).
195 - **Honest LOVO, heuristic vs learned, n=2?n=3:** heuristic mean held-out **0.443?0.544
± 0.278** (Adarsh 0.725,
196 **Kushagra 0.151 capped at any n**, doubles 0.755). Learned prototype mean
**0.377?0.615** (Adarsh F1 0.701/AUC 0.878,
197 **Kushagra 0.049?0.417**/AUC 0.850, **doubles 0.729/AUC 0.914** trained only on the 2
singles). **? The learned model
198 OVERTAKES the heuristic at n=3 (0.615 vs 0.544); at n=2 it trailed (0.377 vs 0.443).**
Crossover predicted by the plan.
199 - **The story (honest):** learned path SCALES with data (+0.24 for one match,
calibration gap closing ? Kushagra
200 0.049?0.417) and GENERALIZES across play type (doubles held-out AUC 0.914); heuristic
is HARD-CAPPED on footage it
201 can't represent (Kushagra ~0.15 at any n). Honest "bar to beat" is now ~0.54 (n=3),
not the fit-on-self 0.73. ?
202 Reusing every labeled video for training validated; bottleneck = data scale +
calibration (n?5), not method.
203 - **Docs updated ? pushed to indexer PR #61** (branch feat/owned-model-study-and-
prototype): n=3 findings + final quality
204 metrics + conclusions in `docs/archives/quality-iterations/2026-06-multivideo-lovo/`;
study + index updated. (PR #60
205 MERGED earlier; #13 MERGED.) ? Prototype speed features NOT yet fps-normalized (30 vs
60fps) yet doubles still
206 generalized (AUC 0.914) ? known refinement.
207 - **OWNED-MODEL: PLAN-ONLY per owner** ? `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
written (DRAFT, uncommitted, for
208 review). User chose "just plan, don't code yet." 4 decisions pending (which Phase-0
milestone to greenlight [rec M0.1
209 corpus spine], WASB C3 stance, ratify Apache-2.0/MIT, confirm collector = corpus
system-of-record). NO owned-model
210 code until greenlight. Momentum = tag more matches (n?5) + I score each.
211 - ?? **NEXT:** owner reviews plan + greenlights a Phase-0 milestone; tag more matches ?
re-run LOVO+prototype (expect
212 learned mean to keep rising). gBaaddy/TestLargeVideo still untagged test videos
(trajectories ready). Commit the plan
213 doc on approval.
214
215 **Checkpoint:** 2026-06-08 (later+2) . ? **Multi-video eval corpus + owned-model
research underway.** User fully tagged
216 Adarsh-and-Avi (40?96 rallies**) + provided **Kushagra** (32 rallies, 4K) for the
TRAINING mix, and **gBaaddy**
217 (new, 1080p, untagged) + TestLargeVideo as held-out TEST videos. Generated full-video
WASB trajectories (single 8GB

```

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218 GPU, serialized via wait-loops): `~/clips/{adarsh_avi_full,kushagra,gbaaddy}_traj.csv`
(+ %TEMP% copies). Scripts:
219 `~/clips/run_wasb_{avi_full,kushagra,gbaaddy}.sh`.
220 - **Adarsh-full eval (96 rallies, IoU>0.5, window [0,1405.8s]):** baseline F1 0.632,
**tuned+gate2 0.727** (P0.80/R0.67,
221 best out-of-box), sweep **ceiling 0.750** (vel0.02/merge1.0/min_dur2.0/gate off, fit-
on-self). vs the opening-10-min
222 (40 rallies): best 0.667 / ceiling 0.742. **Best-config REGION is identical across the
slice and full match**
223 (merge_gap 1.0, min_dur 2.0; velocity irrelevant; gate-off=recall, gate-on=precision)
? early sign tuning may
224 transfer; LOVO is the real test. Per-rally: `output/adarsh_avi_full_vs_human.csv`.
225 - ? **KEY FINDING ? Kushagra: WASB tracks shuttle BETTER than Adarsh (97% of GT rallies
have ?2 net-crossings, med 5,
226 310 pts/rally vs Adarsh 76%) BUT no velocity-windowing config can segment it (per-
video best-fit F1 only 0.163; baseline
227 0.000; mean boundary err ~5s).** Signal is PRESENT; the unsupervised heuristic FAILS ?
likely because Kushagra's shuttle
228 is visible 67.5% (vs Adarsh 40%), i.e. tracked through inter-rally activity ?
boundaries bleed. Root cause is the
229 heuristic class, NOT WASB recall or labels or tuning.
230 - **First cross-video LOVO (`calibrate_local`, Adarsh-full 96 + Kushagra 32, human
labels, manifest
231 `output/human_lovo_manifest.json`):** baseline mean held-out F1 **0.316** ? calibrated
**0.443 ± 0.279** (overfit gap
232 +0.027, tiny). Per-fold: held-out **adarsh 0.722** (cfg tuned on Kushagra ? Adarsh's
own 0.777 ? config TRANSFERS great),
233 held-out **kushagra 0.163** (cfg tuned on Adarsh; = Kushagra's own ceiling).
**Asymmetry is the story:** tuning transfers
234 fine WHERE the heuristic can solve the video; Kushagra is fundamentally unsegmentable
by velocity-windowing (ceiling
235 0.16) regardless of config. The ±0.279 std = method variance, not config. ?
**definitive, data-backed case for the
236 learned OWNED model** (trajectory carries the signal; heuristic can't exploit it).
Kushagra labels+traj = valuable
237 TRAINING data even though the current pipeline can't use them.
238 - ? LOVO honesty: 1 entry per DISTINCT MATCH (group Adarsh-main + Adarsh-last-game as
one unit; calibrate_local has no
239 grouping ? keep matches as separate entries only). gBaaddy traj still inferring (test
video).
240 - ? **PROTOTYPE built (user asked) ? `backend/eval/rally_seq_proto.py` (+3 tests):**
learned per-frame TAS-style
241 segmenter over 6 hand-built WASB-trajectory features (vis-rate, speed, **net-crossing
rate**, spreads) ? pure-Python
242 LogisticRegression, honest LOVO. **Held-out per-frame AUC 0.878 (Adarsh) / 0.841
(Kushagra).** Kushagra: LOVO-threshold
243 F1 0.049 (threshold doesn't transfer at n=2) but **oracle-threshold F1 0.578 = 3.5×
the heuristic's 0.163 ceiling.**
244 ? PROVES the rally signal is LEARNABLE where the heuristic fails; gap = threshold
CALIBRATION not representation; answers
245 the study's "can low-dim WASB features feed a TAS head?" = YES. Seed of Gen-0.
246 - ? **OWNED-MODEL STUDY done (11-agent workflow `wf_49aab888-e10`, 35 sources) ?
`docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`.**
247 3 generations off ONE corpus: Gen-0 ASFormer(MIT)/ActionFormer over WASB features ?
Gen-1 cue fusion ? Gen-2 e2e VLM
248 (Qwen3-VL-4B / Qwen2.5-VL-7B / Gemma-4, ALL Apache-2.0; Gemma-3 REJECTED).
**Platforms: Modal primary (best written
249 no-access), RunPod-Secure fallback, Vertex on bench (India residency but tuning Gemini
? owned weights), 2070S =
250 Gen-0+inference only.** Label-reuse = 3 transcoders, split-by-match. Critic caveats:
**WASB checkpoint provenance C3
251 propagates** (MIT code ? checkpoint provenance), Gemma-4 stay AMBER until raw license
check, **purge Gemini-trained
252 distilled labels** to comply with eval-only policy. DESIGN-ONLY: no impl until P0-P4 +
healthy corpus + owner sign-off.

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253     - ? **DOC CHECKPOINT shipped ? indexer PR #61** (branch feat/owned-model-study-and-
prototype): prototype + study +
254     `docs/archives/quality-iterations/2026-06-multivideo-lovo/` learnings + index README.
Full suite **254 green**.
255     - ? **gBaaddy test trajectory inferring** (started 13:13, ~16 min) ? ready for when user
tags gBaaddy.
256     - ?? **NEXT:** review/merge PR #61. User tags a held-out TEST match (gBaaddy/TestLarge)
? score calibrated config +
257     prototype on truly-unseen footage. **Highest-ROI: grow n?2?5 distinct matches** (small
Roora batch) ? honest LOVO +
258     per-video threshold calibration ? re-run prototype. Then (owner sign-off, post-
scaffold) corpus format + ASFormer Gen-0 A/B.
259
260     **Checkpoint:** 2026-06-08 (later+1) . ? **3 work items shipped (user said "go ahead
with all three").**
261     1. **Tuning sweep** ? added `--sweep` to `backend/eval/eval_partial_labels.py` (+test);
indexer branch `feat/eval-human-labels`
262     (commits 2c950f6 harness, 58d10cd sweep; NOT pushed/PR'd ? offer to PR). On Adarsh-
and-Avi: **F1 0.667?0.742**
263     (velocity 0.03 / merge_gap 1.0 / **min_dur 2.0** / gate OFF; P0.67/R0.83). Dominant
lever = min_window_duration;
264     gate over-prunes this camera; velocity ~irrelevant. ? fit-on-self (one video,
optimistic) ? real number = LOVO.
265     2. **Collector P0 normalizer + positioning doc ? PR #13**
(https://github.com/avidullu/sports-data-collector/pull/13,
266     branch `feat/manual-ingestion-normalizer`, pushed).
`src/cli/normalize_annotations.py` (+test) = standalone
267     pre-import guard: drops blank/?start/**negative-start** rows (else pydantic
validator/safe_float_required abort the
268     WHOLE import) + rennumbers fractional/dup rally_number 1..N by start (else
int(float())) PK-collides + DELETE-INSERT
269     destroys one). Also mm:ss?sec, pipe ending_reason?primary.
`docs/MANUAL_INGESTION_POSITIONING.md` = positioning +
270     schema map + P1/P2 roadmap. **Validated full chain on the REAL file vs a TEMP db**:
normalize 40?40 ? import-local
271     40 ? export 40-interval indexer CSV. Collector suite **124 green** (was 116).
272     3. **Adversarial review** (1 agent) caught a REAL hole pre-merge: negative start_time
was undefined (pydantic
273     `start_time must be non-negative` ? import abort) ? fixed (drop branch + test, commit
1fe1c24) + corrected 2 doc
274     inaccuracies (abort mechanism = pydantic validator not safe_float_required;
ending_reason is free-text not enum).
275     4. **Iteration doc archived + indexer PR'd:** `docs/archives/quality-
iterations/2026-06-human-gt-tuning/` (STATUS
276     COMPLETED; results table, sweep, failure modes, bugs, infra gotchas, reproduce) +
updated the quality-iterations
277     index README + 2026-06-initial seed forward-pointer. **Indexer PR #60 OPEN**
(https://github.com/avidullu/badminton-highlight-indexer/pull/60; branch feat/eval-
human-labels =
279     harness 2c950f6 + sweep 58d10cd + docs; 5 files, +446/-1, full suite 251 green).
280     - **GOTCHA learned:** import-local does NOT copy the video (records local_path); writes
obs report NEXT TO THE VIDEO
281     (clean those up after validation). DB path from `config/config.yaml` `database_path` ?
override via temp config for
282     side-effect-free validation (no `--config`/`--db` CLI flag exists).
283     - ? **BOTH PRs MERGED ? both repos CLEAN on master, no open PRs, branches deleted.**
Indexer **#60** (squash
284     `90fb8b3`: eval_partial_labels harness + sweep + iteration doc). Collector **#13**
(squash `099cf67`: normalize_annotations
285     guard + MANUAL_INGESTION_POSITIONING.md). NOTE: user said "review is merged" while
live API showed both OPEN ? I
286     completed both squash merges + synced.
287     - ?? **NEXT:** after user's 60-min tagging ? full-video re-run
(`backend/eval/eval_partial_labels.py`, window auto =
288     last labeled end) + (recommended) tag BREADTH across 3?4 videos for LOVO. P1 collector

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follow-ups (labeled-window in
289     manifest, maturity ladder needs_review?confirmed?gold) when prioritized.
290
291     **Checkpoint:** 2026-06-08 (later) . ? **FIRST human-GT eval landed + collector manual-
ingestion positioning done.**
292     User did ~10 min of VLC rally-annotator tagging ? partial human CSV `Annotation
Setup/Collect/Adarsh and Avi.rallies.csv`
293     (40 rallies, covers 0?605s of a 23m29s video; schema
`rally_number,start_time,end_time,ending_reason,sport`).
294     - **Built `backend/eval/eval_partial_labels.py` (+test, 3 green) ? committed on branch
`feat/eval-human-labels` (commit
295     2c950f6, NOT pushed/PR'd).** Scores WASB rally segments vs a PARTIAL human CSV,
**restricted to the labeled window
296     [0, last_end]** so correct detections in the un-labeled tail aren't counted as FP (THE
key methodology for partial
297     labels). Reuses calibrate_wasb.make_predict_fn/load_golden_gts + rally_gate + metrics
+ export_wasb_segments.build_comparison.
298     - **Generated WASB trajectory for the labeled window** (first 610s ? 36,564 frames ?
`~/clips/adarsh_avi_traj.csv` +
299     Windows `%TEMP%\adarsh_avi_traj.csv`; WSL script `~/clips/run_wasb_avi.sh`; ~18 min
inference). ? GOTCHA: passing a
300     long multi-statement WSL command via the Bash tool with `run_in_background` MANGLES
`$VAR`/`$()` (empty expansion) ?
301     put the job in a .sh FILE (tr-strip CRLF into WSL) and background `bash file.sh`.
302     - **FIRST trustworthy human-GT numbers (IoU?0.5, 40 rallies in-window):** baseline F1
**0.650** (P/R 0.65/0.65);
303     tuned 0.623 (P0.50/R0.825, over-segments); **tuned+gate2 best F1 0.667**
(P0.79/R0.575). Boundary err ~1s.
304     Per-rally CSV `output/adarsh_avi_vs_human.csv`. **Failure modes:** detector catches
LONG rallies well (IoU 0.7?0.95,
305     tight boundaries) but MISSES short 3?5s rallies (most of the 17 misses) and FRAGMENTS
long ones (golden#1's 34.8s
306     split into 2 sub-fires; most "FPs" are real-rally fragments + one possibly-untagged
early rally @23.5s, NOT
307     hallucinations ? precision is recoverable via merge_gap). gate2 alone hurt recall hard
here (0.65?0.475) ? net-axis
308     est may be off for this camera. Levers: min_window_duration? (short rallies),
merge_gap? (fragmentation).
309     - **Collector manual-ingestion positioning (ultracode 10-agent workflow
`wf_d251ad6c-472`, verdict confidence HIGH):**
310     position sports-data-collector as the **single canonical system-of-record**; indexer
consumes ONLY intervals-only
311     export (manifest.json + per-video start,end CSV). **Architecture ~80% there ? the lean
.rallies.csv imports via
312     `import-local` TODAY with ZERO code changes** (header matches DictReader,
curate.py:232-298). **2 P0 landmines** ?
313     thin pre-import normalizer: (a) blank/?start end_time RAISES and aborts the WHOLE
import (parsing.py:22-28) ? drop bad
314     rows; (b) fractional rally_number (7.5/16.5) truncates via int(float()) and PK-
COLLIDES, DELETE-then-INSERT destroys
315     one (db.py:217) ? renumber 1..N by start_time, keep orig in notes. DON'T widen PK to
float (indexer ignores
316     rally_number). End-state P1/P2: persist `labeled_window_start/end` in manifest (so
indexer auto-restricts instead of
317     inferring max-end ? WRONG when a video is tagged past its last rally); maturity ladder
needs_review?confirmed?gold
318     (? CurationStatus); provenance/annotation_source+timestamp; unify adapters + run Issue
detection on manual imports;
319     freeze intervals-only export contract. Rich `rally_review.csv` (mm:ss, `rally` col,
pipe reasons) NOT directly
320     ingestible ? offline converter (P2, only if it's source of truth ? OPEN question).
Enrichment rides in MANIFEST, never the eval CSV.
321     - ?? **NEXT:** after the 60-min tagging ? re-run full-video (extract full frames + WASB
+ eval, window auto = max end of
322     full labels). For LOVO (learned>baseline, n=2 blocker), recommend tagging BREADTH (3?4

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videos partially) over

323 finishing one. Offer: P0 collector normalizer as a small PR;
`docs/MANUAL\_INGESTION\_POSITIONING.md`.

324

325 ****Checkpoint:**** 2026-06-08 · ? ****rally-annotator v1.3 ? PR #1 MERGED to `main`**** (merge
`d4c682a`, commit `b058262`,
326 5 files +711/?127; local `main` fast-forwarded). Goal: in-plugin Help text +
327 a documented ending-reason taxonomy + UX fixes the user asked for. Files changed in
328 `C:\Users\avidu\Projects\rally-annotator\vlc\rally\_annotator.lua` (full v1.2?v1.3
rewrite), `README.md`,
329 `docs/CSV\_FORMAT.md`, `CHANGELOG.md`, + NEW `docs/ENDING\_REASONS.md`. ****Installed to**
330 `%APPDATA%\vlc\lua\extensions\rally\_annotator.lua` for the user's live smoke test**
(Reload extensions / restart VLC).
331 - ****VLC "About" finding (verified, high-confidence):**** the empty "About ?lua" box the
user screenshotted is VLC's
332 Add-ons `AddonInfoDialog`; its "Lua script" body is a hardcoded constant from the
addon scanner
333 (`modules/misc/addons/fsstorage.c`) ? an extension CANNOT set it. descriptor
`shortdesc`/`description` DO show, but
334 only in the *Active Extensions* tab "More information" dialog. So real, author-
controlled help = IN-DIALOG (Help
335 button + `add\_html`). Enriched the descriptor too.
336 - ****v1.3 plugin changes:**** ending reason now REQUIRED + non-sticky (placeholder `--
choose reason --`; reset after
337 save via dropdown `clear()+rebuild` ? VLC 3.0.x dropdowns have NO `set\_value`, first
value after clear auto-selects);
338 two-step commit (Mark END *arms* ? Save Rally writes; button shows the # it will
write); editable Start/End text
339 fields; Recent-rallies `add\_list` with Edit/Delete (uses `get\_selection()`); Undo last
(cancel edit / clear mark /
340 pop last). Switched to an in-memory rows model + atomic CSV rewrite (tmp+rename,
Windows `.bak` rollback). CSV schema
341 UNCHANGED (`rally\_number,start\_time,end\_time,ending\_reason,sport`); forgiving parser
preserves extra columns.
342 - ****Taxonomy (verified via a 3-agent workflow + adversarial check):**** OUT-of-bounds =
the HITTER's error, forced ONLY
343 if pressured else unforced (NEVER auto-forced, never the opponent's winner); into-net
same; serve-into-net =
344 service_fault; rally net-cord-in = live (often a winner); serve net-cord diverges by
sport; default-to-unforced
345 tie-breaker. One fix the verifier caught and I applied: a badminton serve CAUGHT on
the net = service_fault (BWF Law
346 13.2), NOT a let. Full guide in `docs/ENDING\_REASONS.md`; summary table in README.
347 - ****Verification:**** no Lua interpreter on PATH; adversarial static review (subagent) ?
state machine + VLC 3.0.x API
348 usage clean; fixed the 2 cross-platform bugs it found (POSIX `file:///` path lost
leading `/`; `save\_all` deleted the
349 live CSV before rename). ? STILL NEEDS the user's live VLC click-test (I cannot drive
VLC).
350 - ****Follow-up (same session, also installed):**** (a) RESUME ? on activate the plugin
reloads the video's existing
351 `.rallies.csv` (Recent list + continued numbering); `Refresh` now re-points to
whatever video is playing now and
352 loads its rallies (guarded against reload-during-edit so `edit\_index` can't go stale;
playback *position* is NOT
353 restored). (b) UNDO clarity ? `Undo last` button shows the row it will remove (`Undo
last (#N)`) + a "Last row
354 (what Undo removes)" line in the status panel. README/CHANGELOG updated.
355 - ****Reinstall gotcha (FIXED):**** after merging PR #1 the user hit "plugin not reloading"
? the installed copy was
356 STALE (28251B, an earlier file-copy) vs merged source (28905B). Deleted + reinstalled
the merged file to
357 `%APPDATA%\vlc\lua\extensions\rally\_annotator.lua`, hash-verified byte-for-byte
(28905B). ****VLC 3.x "Reload**
358 extensions" is unreliable for Lua code changes ? a full VLC restart is the reliable

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path** (VLC was not running).
359 Lesson: always reinstall to the VLC dir AFTER the final edit, and verify by SHA-256,
not just byte count.
360 - ? **REAL root cause found (plugin never showed under View) ? fix PR #2 (v1.4) MERGED**
(merge `37a0ec1`,
361 commits `8d42a71` load-fix + `424eb3c` v1.4). The
362 stale file was a red herring; v1.3 NEVER loaded. **VLC scans an extension's
`descriptor()` in a RESTRICTED Lua
363 sandbox that does NOT expose base globals (`ipairs`, etc.).** v1.3 built
`REASON_ID`/`SPORT_ID` with **top-level
364 `ipairs` loops** ? scan-time error `attempt to call global 'ipairs' (a nil value)` ?
VLC silently skipped it. Fix:
365 move the table-building into `activate()` (full env); **NEVER call a global function
at the top level of a VLC Lua
366 extension.** (v1.2 worked only because its `ipairs` calls were all INSIDE functions =
activate-time.)
367 - ? **Reusable diagnosis method:** installed `lua5.1` in WSL ? `luac5.1 -p <file>`
(syntax) + load chunk in an EMPTY
368 env via `setfenv` (simulates the scan sandbox); and launched VLC with `--vv --file-
logging --logfile` + `--no-one-instance`
369 to read the actual `lua debug: Scanning Lua script ?` / `Error loading ?` lines. After
fix the log shows the scan with
370 `capability flags: 0x0` and NO error. WSL has no lua by default; `sudo apt install -y
lua5.1` adds `lua5.1`+`luac5.1`.
371 Also built a reusable **headless `lua5.1` harness** (`%TEMP%\ra_harness.lua`) that
STUBS the vlc dialog API + loads
372 the extension ? drives `activate()`/`show_help()`/`save_rally()` and asserts state ?
catches dialog-logic bugs
373 without VLC.
374 - ? **v1.4 (commit `424eb3c`, in PR #2):** (a) **Help toggle fixed** ? was "stuck"; VLC
didn't repaint the in-place
375 `set_text` on the shared status widget ? now a DEDICATED `add_html` panel removed via
`del_widget` (same trick).
376 (b) **Reason-stickiness fixed** ? selecting `winner` blocked the next rally also being
`winner`; VLC didn't repaint
377 the `clear()`-based dropdown reset ? now the reason RESETS to a savable **`unknown`**
default by RECREATING the
378 dropdown (del_widget+add_dropdown at the same cell; VLC has no `set_value`). Reason no
longer hard-required.
379 (c) **"Next rally #" field** ? resume/insert numbering anywhere; auto-advances to next
FREE number; Undo/Delete
380 re-sync it (no gaps). `unknown` added to README/ENDING_REASONS/CSV_FORMAT.
**Adversarial-review workflow (3 lenses)
381 caught 2 real gap bugs (undo/delete didn't re-sync Next#) ? fixed before commit.** ?
**VLC dialog lesson:**
382 in-place widget updates (`set_text`, dropdown `clear()`+re-add) DON'T reliably repaint
? use `del_widget`+re-add.
383 - ?? **ALL PRs #1/#2/#3/#5 MERGED ? `main` = `e10e543` (v1.5). Issue #4 CLOSED. SLATE
FULLY CLEAN:** only `main`
384 (local==`origin/main`), 0 other branches (all merged ones deleted+pruned
local+remote), 0 open PRs, 0 open issues,
385 working tree clean. Installed VLC file = v1.5.
386 - ? **v1.5 (merged via PR #5, closed issue #4):** (github.com/avidullu/rally-
annotator/pull/5, branch
387 `feat/playback-controls-and-tests`, commit `d8fc66f`). Issue #4 = user's "cognitive
enhancement" request: control
388 VLC playback from the plugin window (no window-switching). **Built: a `Back 5s / Play-
Resume / Pause / Fwd 5s`
389 row.** API **source-verified first** (2-agent workflow vs videolan/vlc-3.0):
extensions DO get `vlc.playlist`
390 (play/pause/status, registered in `extension.c`) + `vlc.var` seek on
`vlc.object.input()`. **Gotcha:** `pause()` is
391 a HARD TOGGLE (`playlist_TogglePause`) ? Pause/Play-Resume are **gated on
`vlc.playlist.status()`** so they never
392 flip wrong. Seek = relative  $\pm 5s$  via `vlc.var.set(input,"time", $\mu s$ )`, clamp 0, nil-

```

```

checked. Layout regrouped (playback
393     row at top; reason cell `(3,4)`?(3,5)`, help row 14?15).
394     - ? **Committed test suite (answers the user's "test coverage?" question):** NEW
`test/dialog_test.lua` (run
395     `lua5.1 test/dialog_test.lua`, exit 0/1) ? stubs VLC dialog+playback API, loads the
real ext, **36 assertions**
396     (reason reset/consecutive/unknown, numbering+collision+undo-resync, help toggle,
**playback gating**, CSV bytes).
397     `test/README.md` + README Tests section. The throwaway `%TEMP%\ra_harness.lua` is now
superseded by this in-repo test.
398     Verified: 36/36 pass, `luac5.1 -p`, empty-env scan `SCAN_OK v1.5`, real VLC `-vv` log
(loads, `0x0`, no error).
399     Installed VLC file = v1.5. ? GUI render + REAL playback still need the user's live
click-through (untestable headless).
400     - ?? **NEXT:** nothing pending in the repo. User restarts VLC ? live-test playback +
layout x 5 sports; iterate on
401     feedback. Cheap future ideas floated: tweak seek step (5s?3s / add ?2s), per-sport
keyboard shortcuts (space=play/pause,
402     arrows=seek, S/E=mark). Run tests anytime: `lua5.1 test/dialog_test.lua` (36
assertions).
403
404     **Checkpoint:** 2026-06-07 (FINAL, session wrap) . ? **ALL MERGED ? observability
feature fully shipped; both repos
405     CLEAN.** Zero open PRs across all 3 repos (indexer/collector/obsreport); both repos on
`master`, 0 uncommitted, no
406     stray branches. Merged this session: indexer **41/#42/#43/#44**, collector
**10/#11/#12**, obsreport on GitHub
407     (private, v0.1.3).
408     - **44 (last one)** = docs+tooling handoff: **`docs/CODE_MAP.md` REGENERATED** for the
#40 layered layout + config/
409     reporting subsystems (build-code-map workflow, 0 broken anchors, 325 lines) .
**`run_all_tests.py`** at repo root
410     (one-command full suite; README points to it) . Grok coordination notes in
`docs/BACKLOG.md` + `LOW_REGRET_MOVES_PLAN.md`.
411     - ?? **TWO config-migration bugs flagged for Grok** (in BACKLOG + CODE_MAP §2.0 gotchas,
found by the regen, NOT fixed ?
412     Grok's config domain): (1) `/api/compile` ignores `--output-dir` (`OutputConfig` has
no `output_dir` field ?
413     always "output"); (2) `--output-dir` lives only on dynamic
`global_config.runtime_overrides`. Fix: add
414     `output_dir` to a config/runtime model + read consistently.
415     - **Suites:** indexer 231, collector 116, obsreport 74 ? all green. Run via `python
run_all_tests.py`.
416     - ?? **NOTHING PENDING for the observability work.** Optional future (not started): HTML
customer renderer; real
417     AI-provider timing (via Grok item #7); pin `obsreport` git+ssh tag once SSH/CI access
exists; extend report with
418     audio-readiness (Grok item #8). Original rally-detection modeling still PARKED pending
golden labels.
419
420
421
422     **Checkpoint:** 2026-06-07 (latest+2) . ? **FEATURE COMPLETE ? observability in both
tools, merged; Phase D done.**
423     MERGED to master: obsreport (GitHub private, v0.1.3) . indexer **42** (config fix) +
**41** (observability) .
424     collector **10** (parser fix) + **11** (observability). **Debuggability eval done**
(15-agent Workflow: all 7
425     broken-run scenarios self-debuggable; fresh readers given only README/FAQ landed on the
root cause + cited the FAQ).
426     Refinements applied as follow-up PRs (OPEN, green, awaiting owner merge): indexer
**43** (README Q7 lists
427     wasb_hybrid/tracknet_hybrid; new fps Q14 + repoint fps_fallback off Q5; numeric fps
fallback; motion synthetic at
428     data level; silent_provider_noop segmenter?handoff + .env path; blur remediation; **faq-

```


resolution golden test** ?

429 231 green) and collector ****#12**** (frame-level evidence on delivery flags; review-only framing for

430 possible_missed_rally/shot_density; ****faq-resolution golden test**** ? 116 green). Eyeballed reports via rendered

431 HTML screenshot + customer summaries (clean, audience-appropriate). Eval artifacts: ``%TEMP%\obseval\``.

432 ?? ****NEXT** (when owner is back):** merge #43 + #12 (both green, disjoint). Then the per-video observability feature is

433 fully shipped across both tools + the shared obsreport lib. Optional future: HTML customer report renderer (obsreport);

434 thread real AI-provider timing through the segmenters (currently 'not_captured'); pin obsreport via git+ssh once SSH/CI access is set up.

435

436 ****Checkpoint:**** 2026-06-07 (latest+1) . ? ****INDEXER OBSERVABILITY MERGED.**** Both indexer PRs merged to master:

437 ****#42**** (config fix ``AppConfig.get`` nested?dict; review nits addressed: dynamic section scan + shim comment + tests)

438 and ****#41**** (per-video observability reports) ? master ``98bb1d9``, ****229 tests green****, feat branch deleted. Full

439 pipeline now sound end-to-end (validators run via #42; every run emits report via #41). obsreport on GitHub (private,

440 tags `v0.1.0?v0.1.3`). ****Collector observability PR #11 STILL OPEN**** (rebased on #10, 113 green).

441 ?? ****NEXT:** Phase D refinements (follow-up PRs) + golden tests.** Eval verdict: all 7 broken-run scenarios

442 self-debuggable. Apply: (HIGH) repoint ``fps_fallback_used`` `faq_ref` off README#Q5 (keyframes) to a NEW fps FAQ; add

443 ``wasb_hybrid`/`tracknet_hybrid`` to README Q7 segmenter list. (MINOR) numeric fps fallback ``30`` (fallback); motion

444 "synthetic" surfaced at data level; self-remediating blur line (name ``validation.min_blur_threshold``); collector

445 overlap frame-level detail; ``possible_missed_rally`` review-only framing. Add golden tests: `faq-ref-resolution` (every

446 flag's cited anchor exists in its doc) + expected flag-set per fixture, in BOTH repos. Indexer refinements ? new

447 branch off master; collector ? push to #11. Eval artifacts in ``%TEMP%\obseval\`` (manifest.json + reports/ + `gen_*.py`).

448

449 ****Checkpoint:**** 2026-06-07 (latest) . ? ****RE-SYNCED** onto parallel work + found/fixed a config runtime bug.**

450 Indexer master advanced: ****#39** (typed config backend/config/)** + ****#40** (layered directory restructure:

451 backend/pipeline/{validators,segmenters,stitcher,detectors,audio}, backend/infrastructure/database.py,

452 backend/api/static/)**. Rebased ****observability PR #41** onto #40** (git rename-detection merged my main.py reporting

453 onto the new imports; only repointed 2 test imports) ? ****228 tests green****; #40 also fixed the 2 config tests I'd

454 flagged. Collector observability ****PR #11** rebased onto #10** (parser fix merged) ? ****113 green****. obsreport ****74 green****.

455 - ? ****FOUND + FIXED** a real config runtime bug ? indexer PR #42 (``fix/appconfig-nested-get``).** The typed AppConfig is

456 passed to validators/segmenters which use legacy chained ``config.get("validation",{ }).get("k")``; AppConfig.get

457 returned the nested `*model*` (no ``get``) ? ****all 4 validators** threw ``'ValidationConfig' object has no attribute 'get'``

458 on every real ``python -m backend.main`` run** (tests passed because they inject dicts). Fix: AppConfig.get returns

459 nested sections via ``model_dump`` (deep dict-compatible); +regression test; ****215 green****; CLI smoke now runs all

460 validators. ****Surfaced BY the observability report**** (nice dogfood). Owner approved fixing it (its own small PR #42).

461 - ****Open PRs now:**** indexer ****#41**** (observability, green), indexer ****#42**** (config fix, green), collector ****#11****

```

462      (observability, green). obsreport on GitHub (private, tags v0.1.0?v0.1.3). Suggested
merge order: ##42 ? #41
463      (so #41's runtime path is unbroken), and ##11 any time.
464      - ?? STILL PENDING: Phase D ? debuggability eval Workflow was running (fresh-reader
agents over 7 broken-run
465      reports); apply synthesized wording refinements + render a report to an image to
eyeball + add golden faq-resolution
466      tests, then push to #41/#11.
467
468      **Checkpoint:** 2026-06-07 (later) . ? **NEW FEATURE IN PROGRESS ? per-video
OBSERVABILITY REPORTS** across
469      BOTH tools (indexer + collector) + a NEW shared lib. Approved plan:
470      `C:\Users\avidu\claudu\plans\drifting-snuggling-squirrel.md` (READ IT to resume).
Decisions: always-on,
471      reports written NEXT TO THE VIDEO by default (override via `report.output_dir`); BOTH
tools emit a technical
472      report (debuggable by a README+FAQ reader) AND a customer-friendly summary (strip
internals); shared schema via
473      a **new private repo `sports-obsreport` (pkg `obsreport`) ? a reusable "report config
language."
474      **Research:** Soda/SodaCL = ELv2 (disqualified, not OSS);
GE/Evidently/ydata/Frictionless license-safe but
475      wrong shape ? built a small lib on permissive primitives (pydantic/Jinja2/PyYAML).

```

31. user (2026-06-10T05:53:44.328Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

32. user (2026-06-10T05:53:53.275Z)

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit
`wf_2a64bc88-2a8` over indexer+collector:
13     engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors
completed+verified, 5 rate-limited).
14     - ? **CRITICAL ? master is BROKEN since 7fbbbe0 (2026-06-07 print?logger sweep):**
15     `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines
(100/104/156/160/181/191/213)** ? SyntaxError ?

```

33. user (2026-06-10T05:54:04.304Z)

The file c:\Users\avidu\claudu\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

34. assistant (2026-06-10T05:54:19.935Z)

****Done ? [PR #63](https://github.com/avidullu/badminton-highlight-indexer/pull/63)** is open****** with the fix, the regression guard, and a fully green suite.

****The fix**** ([compiler.py](backend/pipeline/stitcher/compiler.py)): restored the quotes on all 7 mangled lines (100, 104, 156, 160, 181, 191, 213) and added the missing ` ` on line 168 (`f"\nSUMMARY` ? `f"\nSUMMARY`)). I verified the message texts against the pre-conversion file

at `7fbbeb0` ? they're unchanged. One small judgment call: line 191 ("Stitching clips together...") has no placeholders, so I restored it as a plain string rather than a pointless `f"`, matching the file's existing convention (e.g. line 82). Nothing else in the file was touched.

****Verification:****

- `python -m compileall backend` ? clean
- `python -c "import backend.main"` ? OK (FastAPI server and `/api/compile` importable again)
- Negative check: master's version of the file fails compilation with `SyntaxError: invalid decimal literal` at line 100, confirming the guard catches exactly this bug
- `python run\_all\_tests.py` ? ****256 passed**** (254 existing + 2 new), offline/mock

****The regression guard**** (tests/test_import_smoke.py) has two tests: a `py\_compile` sweep over every file in `backend/` (imports nothing, so it also runs in the GPU-less dev container where cv2/torch/numpy are absent), and an import of `backend.main` in a subprocess with any *missing* heavy deps stubbed via `MagicMock` ? subprocess isolation keeps the stubs from leaking into the rest of the pytest session.

Branch is `fix/compiler-fstring-syntax` off updated master, one commit, one PR per the repo conventions. I also added a checkpoint to the memory `current-state.md` noting the fix is up (this was the audit's flagged critical / fix-chip item). One thing left for you: merge #63 ? and note the audit's broader findings (coverage inverted vs. production risk, no CI) are still open; this PR only closes the import/syntax gap.

35. user (2026-06-10T05:54:48.641Z)

does this also add a github CI test for ensuring its protected in the future?

36. assistant (2026-06-10T05:55:30.448Z)

No ? PR #63 only added the pytest guard; the repo has no CI at all (that was one of the audit findings), so the guard only fires when someone runs the tests locally. Let me close that gap now by adding a GitHub Actions workflow to the same PR. First I need to see what the test suite actually needs to run on a bare runner:

37. user (2026-06-10T05:55:31.491Z)

```
pytest.ini
requirements.txt
setup.py
```

38. user (2026-06-10T05:55:32.819Z)

```
tests\test_eval_classifier.py:1:import numpy as np
tests\test_hit_detector.py:2:import numpy as np
tests\test_rally_seq_proto.py:9:import numpy as np
tests\test_tracknet_hybrid.py:5:import cv2
tests\test_wasb_cache.py:11:import numpy as np
tests\test_yolo_hybrid.py:8:    import cv2
tests\test_yolo_hybrid.py:279:    import numpy as np
```

39. user (2026-06-10T05:55:40.360Z)

```
1      fastapi>=0.111.0
2      uvicorn>=0.30.1
3      opencv-python>=4.9.0.80
4      numpy>=1.26.4
5      pydantic>=2.7.1
6      jinja2>=3.1.4
7      python-dotenv>=1.0.0
8      google-genai>=2.4.0
9      torch>=2.12.0
10     torchvision>=0.27.0
11     supervision>=0.21.0,<0.30.0
```

```

12
13     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
14     # no-op if absent). Shared, permissively-licensed report toolkit. Pin a tag once
15     # the private repo is published, e.g.:
16     #   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
17     # For local development install it editable from a sibling checkout:
18     #   pip install -e ../sports-obsreport --no-build-isolation
19

```

40. user (2026-06-10T05:55:40.834Z)

```

1     [pytest]
2     # When `pytest` is invoked without an explicit path, only collect from tests/.
3     # scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.)
4     # and should never run as part of the regular test suite.
5     testpaths = tests
6

```

41. user (2026-06-10T05:55:42.121Z)

```

backend\pipeline\detectors\wasb_infer.py:212:     import torch
backend\pipeline\detectors\wasb_infer.py:213:     from torch.utils.data import DataLoader
backend\pipeline\segmenters\yolo_hybrid.py:17:import torch
backend\pipeline\segmenters\yolo_hybrid.py:73:     from torchvision.models import detection
as tv_detection

```

42. user (2026-06-10T05:57:17.042Z)

```

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run:
C:\Users\avidu\AppData\Local\Temp\ci_smoke_venv\Scripts\python.exe -m pip install --upgrade
pip

```

43. user (2026-06-10T05:57:25.043Z)

```

Exit code 1
>     assert result.returncode == 0, (
        "import backend.main failed:\n" + result.stderr.strip()
    )
E     AssertionError: import backend.main failed:
E     Traceback (most recent call last):
E       File "<string>", line 6, in <module>
E         import backend.main
E       File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py", line 27,
in <module>
E         from backend.pipeline.segmenters.base import segmenter_registry, BasePlaySegmenter
E       File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\__init__.py", line 3, in <module>
E         from backend.pipeline.segmenters.gemini import GeminiPlaySegmenter
E       File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py", line 12, in <module>
E         from backend.providers import provider_registry
E       File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\__init__.py", line 3, in <module>
E         from backend.providers.gemini import GeminiProvider
E       File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py", line 6, in <module>
E         from google import genai
E       ModuleNotFoundError: No module named 'google'
E     assert 1 == 0
E     + where 1 = CompletedProcess(args=['C:\Users\avidu\AppData\Local\Temp\ci_smoke_v
env\Scripts\python.exe', '-c', "import importlib.util, sys\nfrom unittest import mock\nfor
name in ('cv2', 'torch', 'numpy'):\n    if importlib.util.find_spec(name) is None:\n

```

```

sys.modules[name] = mock.MagicMock()\nimport backend.main\n"], returncode=1, stdout='',
stderr='Traceback (most recent call last):\n File "<string>", line 6, in <module>\n    import
backend.main\n File "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\main.py", line 27, in <module>\n    from backend.pipeline.segmenters.base
import segmenter_registry, BasePlaySegmenter\n File "C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\pipeline\\segmenters\\__init__.py", line 3, in <module>\n    from
backend.pipeline.segmenters.gemini import GeminiPlaySegmenter\n File
"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\pipeline\\segmenters\\gemini.py", line 12, in <module>\n    from
backend.providers import provider_registry\n File "C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\providers\\__init__.py", line 3, in <module>\n    from
backend.providers.gemini import GeminiProvider\n File "C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\providers\\gemini.py", line 6, in <module>\n    from google import
genai\nModuleNotFoundError: No module named \'google\'\n').returncode

```

```

tests\test_import_smoke.py:53: AssertionError
===== short test summary info =====
FAILED tests/test_import_smoke.py::test_backend_main_imports - AssertionError...
===== 1 failed, 1 passed in 0.99s =====

```

44. assistant (2026-06-10T05:57:37.745Z)

The venv experiment caught a real dependency: the import chain needs `google-genai`. Adding it and re-running:

45. user (2026-06-10T05:58:02.246Z)

```

[notice] To update, run:
C:\Users\avidu\AppData\Local\Temp\ci_smoke_venv\Scripts\python.exe -m pip install --upgrade
pip
..
2 passed in 2.07s

```

[100%]

46. assistant (2026-06-10T05:58:39.109Z)

name: CI

```

on:
  push:
    branches: [master]
  pull_request:

```

```

permissions:
  contents: read

```

```

concurrency:
  group: ci-${{ github.ref }}
  cancel-in-progress: true

```

jobs:

```

# Fast guard against the 7fbb0 class of breakage: a SyntaxError or
# module-level import error shipping while the suite stays green.
# Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
# on machines without the GPU stack (test_import_smoke stubs what's absent).
import-smoke:

```

```

  name: Syntax sweep + import smoke (no GPU stack)
  runs-on: ubuntu-latest
  steps:

```

```

    - uses: actions/checkout@v4
    - uses: actions/setup-python@v5
      with:

```

```

        python-version: "3.13"

```

```

    - name: Install minimal deps

```

```

      run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai

```

```

- name: Run import smoke tests
  run: python -m pytest tests/test_import_smoke.py -v

full-suite:
  name: Full test suite (offline, mocked)
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - uses: actions/setup-python@v5
      with:
        python-version: "3.13"
        cache: pip
    - name: System libs for opencv-python
      run: |
        sudo apt-get update
        sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y libgl1
        libglib2.0-0
    - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
      run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
    - name: Install requirements
      run: python -m pip install -r requirements.txt pytest
    # obsreport (private, soft dep) is absent here: reporting tests skip via
    # pytest.importorskip ? run_all_tests.py prints which.
    - name: Run full suite
      run: python run_all_tests.py

```

47. user (2026-06-10T05:58:39.129Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\.github\workflows\ci.yml (file state is current in your context ? no need to Read it back)

48. user (2026-06-10T05:58:53.600Z)

warning: in the working copy of '.github/workflows/ci.yml', LF will be replaced by CRLF the next time Git touches it

[fix/compiler-fstring-syntax be42266] ci: add GitHub Actions workflow (import smoke + full suite)

1 file changed, 53 insertions(+)

create mode 100644 .github/workflows/ci.yml

To https://github.com/avidullu/badminton-highlight-indexer.git

e14df4e..be42266 fix/compiler-fstring-syntax -> fix/compiler-fstring-syntax

49. user (2026-06-10T05:59:01.512Z)

Command running in background with ID: bhtv7c684. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\2139f383-5578-4e22-831e-8859ad4f69f2\tasks\bhtv7c684.output. You will be notified when it completes. To check interim output, use Read on that file path.

50. assistant (2026-06-10T05:59:07.520Z)

While that runs, updating the PR description to cover the CI addition:

51. user (2026-06-10T05:59:20.738Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/63>

52. assistant (2026-06-10T05:59:27.320Z)

PR body updated. Checking how the CI run is progressing:

53. user (2026-06-10T05:59:29.092Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

54. assistant (2026-06-10T05:59:35.022Z)

The watcher buffers its output until done. Quick direct status check instead:

55. user (2026-06-10T05:59:38.252Z)

```
in_progress      fix(stitcher): restore f-string quotes dropped in logging conversion
(SyntaxError on master)  CI      fix/compiler-fstring-syntax      pull_request      27256500637
38s      2026-06-10T05:58:59Z
```

56. assistant (2026-06-10T05:59:59.525Z)

The existing background watcher (`gh pr checks --watch`) already covers this ? it exits on completion either way, so no extra monitor needed. CI run [27256500637](https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27256500637) is in progress; I'll report as soon as both jobs finish.

57. user (2026-06-10T06:00:39.673Z)

please go ahead and merge the PR if CI is green and really appreciate you finding and fixing it!

58. assistant (2026-06-10T06:00:45.964Z)

Thank you! Checking CI status now ? I'll merge once both jobs are green:

59. user (2026-06-10T06:00:50.202Z)

```
Full test suite (offline, mocked)      pass      1m29s      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27256500637/job/80491884375
Syntax sweep + import smoke (no GPU stack)      pass      27s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27256500637/job/80491884426
```

60. user (2026-06-10T06:01:14.058Z)

```
From https://github.com/avidullu/badminton-highlight-indexer
* branch      master      -> FETCH_HEAD
a5477c2..13d4573 master      -> origin/master
Updating a5477c2..13d4573
Fast-forward
 .github/workflows/ci.yml      | 53 ++++++
 backend/pipeline/stitcher/compiler.py | 16 +++++
 tests/test_import_smoke.py      | 55 ++++++
 3 files changed, 116 insertions(+), 8 deletions(-)
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 tests/test_import_smoke.py
```

61. assistant (2026-06-10T06:01:21.225Z)

Merged ? master fast-forwarded to `13d4573`. Verifying the final state and the post-merge CI run on master:

62. user (2026-06-10T06:01:31.889Z)

```
master
13d4573 fix(stitcher): restore f-string quotes dropped in logging conversion (SyntaxError on
master) (#63)
a5477c2 docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball)
(#62)
backend.main import OK on master
in_progress      fix(stitcher): restore f-string quotes dropped in logging conversion ?
```

CI master push 27256594049 18s 2026-06-10T06:01:12Z

63. user (2026-06-10T06:01:53.735Z)

The file c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

64. user (2026-06-10T06:02:03.901Z)

in_progress fix(stitcher): restore f-string quotes dropped in logging conversion ?
CI master push 27256594049 51s 2026-06-10T06:01:12Z

65. user (2026-06-10T06:03:12.093Z)

master CI run GREEN

66. assistant (2026-06-10T06:03:25.809Z)

****Merged and verified.**** [PR #63](https://github.com/avidullu/badminton-highlight-indexer/pull/63) is squash-merged as `13d4573`, the branch is deleted, and master is healthy again ? `import backend.main` works, so the FastAPI server and `/api/compile` are back.

CI confirmation, both rounds:
- ****On the PR:**** import-smoke passed in 27s, full suite passed in 1m29s ? green on the very first run, no iteration needed.
- ****On master post-merge:**** the push-triggered run also finished green, so the workflow is now live as the standing guard ? every future PR and master push runs the syntax sweep, the `backend.main` import check, and the full 256-test suite.

I also updated the memory checkpoint: the audit's "no CI exists at all" finding is now closed, and master's broken state is resolved. The one related item still open from the audit is the coverage inversion (e.g. `wasb\_hybrid` and the concrete validators have little to no test coverage) ? CI can only guard what the suite covers, so that's worth its own work item when you want it.

And thank you ? glad it was caught before anything tried to serve `/api/compile` off master!

67. user (2026-06-10T09:30:01.241Z)

Can you prepare a report for Roora based on the data we ingested yesterday? I would like to preview it before you make it into a pdf. Additionally, can you give me a quick table of the rally start and end of the pickleball and tabletennis matches that they annotated, so I can cross check their correctness. Did you find any anomalies in their data? please show the data first and once I cross check then the report can follow

68. assistant (2026-06-10T09:30:25.338Z)

Pulling up the ingested Roora data first ? the converted CSVs live in `%TEMP%\roora\` (with the raw CVAT XMLs alongside). Let me verify they're still there and read them:

69. user (2026-06-10T09:30:28.587Z)

FullName	
Length LastWriteTime	
-----	-----

C:\Users\avidu\AppData\Local\Temp\roora\annotations.report.json	2007 6/9/2026 6:05:01 PM
C:\Users\avidu\AppData\Local\Temp\roora\annotations.report.md	662 6/9/2026 6:05:01 PM
C:\Users\avidu\AppData\Local\Temp\roora\annotations.summary.md	478 6/9/2026 6:05:01 PM
C:\Users\avidu\AppData\Local\Temp\roora\pickleball_rallies.csv	454 6/9/2026 6:05:01 PM
C:\Users\avidu\AppData\Local\Temp\roora\tabletennis_rallies.csv	419 6/9/2026 6:05:01 PM
C:\Users\avidu\AppData\Local\Temp\roora\pb\annotations.xml	50145 6/9/2026 6:05:00 PM
C:\Users\avidu\AppData\Local\Temp\roora\tt\annotations.xml	46678 6/9/2026 6:05:00 PM

70. assistant (2026-06-10T09:30:34.229Z)

All there. Reading the CSVs plus the conversion reports:

71. user (2026-06-10T09:30:34.720Z)

```
1 rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot
_outcome,score_before,score_after,notes
2 1,0:00.000,0:01.001,0.000,1.001,,service_fault,0,,,
3 2,0:11.011,0:18.519,11.011,18.519,,unforced_error,0,,,
4 3,0:31.532,0:34.535,31.532,34.535,,unforced_error,0,,,
5 4,0:55.556,0:58.058,55.556,58.058,,unforced_error,0,,,
6 5,1:08.068,1:15.075,68.068,75.075,,winner,0,,,
7 6,1:29.089,1:33.594,89.089,93.594,,unforced_error,0,,,
8
```

72. user (2026-06-10T09:30:35.181Z)

```
1 rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot
_outcome,score_before,score_after,notes
2 1,0:02.503,0:06.006,2.503,6.006,,other,,,,out of table
3 2,0:11.011,0:12.012,11.011,12.012,,other,,,,net
4 3,0:16.016,0:19.520,16.016,19.520,,other,,,,
5 4,0:28.529,0:33.033,28.529,33.033,,other,,,,
6 5,0:37.037,0:45.546,37.037,45.546,,winner,,,,
7 6,0:51.051,0:55.556,51.051,55.556,,other,,,,
8
```

73. user (2026-06-10T09:30:35.971Z)

```
1 # Sports Data Collector ? Delivery / Ingestion Report
2
3 ## Run
4
5 | Field | Value |
6 | --- | --- |
7 | Tool | sports-data-collector |
8 | Video id | annotations (human_id) |
9 | Generated | 2026-06-09T12:35:01Z |
10 | Schema | 1.0 |
11
12 ## Video
13
14 | Field | Value |
15 | --- | --- |
16 | Duration | 58.1s |
17 | FPS | 59.94 fps (declared) |
18
19 ## Processing stages
20
21 **parse** ? `ok` (0.0s)
22 - rallies_parsed: 6
23 - _source_format_: cvat
24 - _fps_: 59.94
25 - _sampling_step_: 30
26
27 **validation** ? `ok` (0.0s)
28 - issues_blocker: 0
29 - issues_high: 0
30 - issues_medium: 0
31 - issues_low: 0
32 - n_blockers: 0
33 - _gate_: PASS
34
```

```
35     ## Findings & issues
36
37     _No findings._
38
39     ## Verdict
40
41     **PASS**
42
```

74. user (2026-06-10T09:30:36.106Z)

```
1     # Sports Data Collector ? Delivery / Ingestion Report (summary)
2
3     ## Run
4
5     | Field | Value |
6     | --- | --- |
7     | Video | annotations |
8     | Report generated | 2026-06-09T12:35:01Z |
9
10    ## Video
11
12    | Field | Value |
13    | --- | --- |
14    | Duration | 58.1s |
15    | Frame rate | 59.94 fps |
16
17    ## Rally summary
18
19    - **6** rallies
20    - **25.5s** of play
21    - **44%** of the video is rallies
22
23    ## Things to review
24
25    _Nothing needs your attention._
26
27    ## Verdict
28
29    This delivery passed all checks and is ready to import.
30
```

75. user (2026-06-10T09:30:56.781Z)

```
PB boxes: 57
--- attribute names (distinct) ---
end time
ending_reason
last_shot_outcome
rally_number
score_after(A-B)
score_before(A-B)
shot_count
start time
--- first box of rally 1 and 2, full attrs ---
frame= rally=1
    rally_number = '1'
    start time = '0'
    end time = '0.06'
    shot_count = '1'
    score_before(A-B) = ''
    score_after(A-B) = ''
    ending_reason = 'service_fault'
    last_shot_outcome = '0'
frame= rally=2
```

```
rally_number = '2'
start time = '0.73'
end time = '1.23'
shot_count = '7'
score_before(A-B) = ''
score_after(A-B) = ''
ending_reason = 'unforced_error'
last_shot_outcome = '0'
```

76. user (2026-06-10T09:31:33.308Z)

IMAGE MODE

```
<image id="0" name="frame_000000" width="3840" height="2160"><box label="rally" source="manual"
occluded="0" xtl="2176.62" ytl="956.82" xbr="2257.72" ybr="1121.76" z_order="0"><attribute
name="rally_number">1</attribute><attribute name="start time">0</attribute><attribute name="end
time">0.06</attribute><attribute name="shot_count">1</attribute><attribute
name="score_before(A-B)"></attribute><attribute name="score_after(A-B)"></attribute><attribute
name="ending_reason">service_fault</attribute><attribute
name="last_shot_outcome">0</attribute></box></image>
```

77. assistant (2026-06-10T09:31:39.715Z)

Image-mode CVAT with frame numbers in the image names. Running the full per-rally extraction and consistency scan on both sports:

78. user (2026-06-10T09:31:51.033Z)

```
=== pb ? 57 boxes, rallies 1,2,3,4,5,6
rally 1: boxes=3 frames=0-60 -> 0s-1.001s | vendor='0'-'0.06' shots='1' score='->'
reason='service_fault' outcome='0' notes=''
rally 2: boxes=16 frames=660-1110 -> 11.011s-18.519s | vendor='0.73'-'1.23' shots='7'
score='->' reason='unforced_error' outcome='0' notes=''
rally 3: boxes=7 frames=1890-2070 -> 31.532s-34.535s | vendor='2.1'-'2.3' shots='2' score='->'
reason='unforced_error' outcome='0' notes=''
rally 4: boxes=6 frames=3330-3480 -> 55.556s-58.058s | vendor='3.7'-'3.86' shots='2'
score='->' reason='unforced_error' outcome='0' notes=''
rally 5: boxes=15 frames=4080-4500 -> 68.068s-75.075s | vendor='4.53'-'5' shots='5' score='->'
reason='winner' outcome='0' notes=''
rally 6: boxes=10 frames=5340-5610 -> 89.089s-93.594s | vendor='5.93'-'6.23' shots='3'
score='->' reason='unforced_error' outcome='0' notes=''
box frame spacing distinct: 30,600,780,840,1260 (30 = in-rally sampling; larger = between
rallies)
=== tt ? 57 boxes, rallies 1,2,3,4,5,6
rally 1: boxes=8 frames=150-360 -> 2.503s-6.006s | vendor='0.16'-'0.4' shots='5' score='->'
reason='other' outcome='' notes=''
rally 2: boxes=3 frames=660-720 -> 11.011s-12.012s | vendor='0.73'-'0.8' shots='2' score='->'
reason='other' outcome='' notes=''
rally 3: boxes=8 frames=960-1170 -> 16.016s-19.52s | vendor='1.06'-'1.3' shots='4' score='->'
reason='other' outcome='' notes=''
rally 4: boxes=10 frames=1710-1980 -> 28.529s-33.033s | vendor='1.9'-'2.2' shots='6'
score='->' reason='other' outcome='' notes=''
rally 5: boxes=18 frames=2220-2730 -> 37.037s-45.546s | vendor='2.46'-'3.03' shots='12'
score='->' reason='winner' outcome='' notes='' ? INTRA-RALLY MISMATCH: reason=[other|winner]
rally 6: boxes=10 frames=3060-3330 -> 51.051s-55.556s | vendor='3.4'-'3.7' shots='7'
score='->' reason='other' outcome='' notes=''
box frame spacing distinct: 30,240,300,330,540 (30 = in-rally sampling; larger = between
rallies)
```

79. user (2026-06-10T09:32:52.964Z)

TT distinct attr names:
end time
ending_reason
last_shot_outcome

```
note
rally_number
score_after(A-B)
score_before(A-B)
shot_count
start time
```

non-empty 'note'-ish values:

```
note = 'net'
note = 'out of table'
```

True

80. assistant (2026-06-10T09:33:13.374Z)

I have the full picture now. Flagging the converter bug for the collector repo first, then here's your cross-check data:

81. user (2026-06-10T09:33:20.615Z)

Noted (position 1, task_id: task_c55f0be6). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_c55f0be6 "Fix rally_cvat ATTR_MAP dropping shot_count". Continue your current work.

82. assistant (2026-06-10T09:33:49.911Z)

Cross-check tables

Times are recomputed from the annotated frame ranges at the true 59.94 fps (the authoritative method ? see anomaly #2 about the vendor's own time fields). Shot counts are pulled straight from their XML, since our CSV dropped them (anomaly #1).

Pickleball ? `Video 2 - Pickleball.mp4` (97 s, 4K @ 59.94 fps)

#	Start	End	Duration	Shots	Ending reason	Last-shot outcome
1	0:00.000	0:01.001	1.0 s	1	service_fault	`0` ?
2	0:11.011	0:18.519	7.5 s	7	unforced_error	`0` ?
3	0:31.532	0:34.535	3.0 s	2	unforced_error	`0` ?
4	0:55.556	0:58.058	2.5 s	2	unforced_error	`0` ?
5	1:08.068	1:15.075	7.0 s	5	winner	`0` ?
6	1:29.089	1:33.594	4.5 s	3	unforced_error	`0` ?

Table tennis ? `Video 3 - Tabletennis.mp4` (58 s, 4K @ 59.94 fps)

#	Start	End	Duration	Shots	Ending reason	Note
1	0:02.503	0:06.006	3.5 s	5	other ?	"out of table"
2	0:11.011	0:12.012	1.0 s	2	other ?	"net"
3	0:16.016	0:19.520	3.5 s	4	other ?	?
4	0:28.529	0:33.033	4.5 s	6	other ?	?
5	0:37.037	0:45.546	8.5 s	12	winner ?	?
6	0:51.051	0:55.556	4.5 s	7	other ?	?

Anomalies found

On our side (not Roora's fault):

1. **Our converter drops `shot\_count`.** Their XML has it filled for all 12 rallies, but `rally\_cvat.py`'s attribute mapping misses the name, so the CSV column is empty. I've flagged a fix chip for the collector repo (the table above reads the counts directly from XML, so you can still cross-check them).

****In Roora's data:****

2. ****Their `start time`/`end time` fields are wrong by ~4x.**** Decoded the format: every value equals ``frame ÷ 900``, i.e. decimal minutes computed assuming 15 fps on a 59.94 fps video (e.g. pickleball rally 2 truly starts at 11.0 s but their field says 0.73 "minutes" = 43.8 s). They're internally consistent, just not wall-clock ? our converter already ignores them in favor of frame ranges, but Roora should fix or drop the field.
3. ****Pickleball `last\_shot\_outcome` is the literal string `0` on all 6 rallies**** ? off-vocabulary (expected ``in`/`n_a`/`net`/`out``); looks like an unedited default. Table tennis left it blank entirely.
4. ****Table-tennis `ending\_reason` is `other` on 5 of 6 rallies****, with the real reason shoved into the free-text note on only 2 of those ("out of table", "net"). The taxonomy isn't being used; rallies 3, 4, 6 have no usable ending reason at all.
5. ****TT rally 5 is internally inconsistent****: its 18 boxes disagree on ending_reason (``other`` vs ``winner``). Our converter happened to pick ``winner`` ? plausible given 12 shots, but it's annotator noise.
6. ****Scores are empty everywhere**** ? ``score_before/after(A-B)`` blank on every box in both sports.
7. ****Pickleball rally 1 starts at frame 0 (0:00.000)**** ? a 1-shot, 1-second service fault beginning on the video's very first frame. Possibly real, but worth checking whether the clip starts mid-serve (clipped rally).

****Quality positives, for fairness:**** box sampling is complete and disciplined (exactly every 30 frames inside every rally, no gaps), rally numbers are sequential with no overlaps, and shot counts are plausible against durations (e.g. 12 shots in 8.5 s of TT).

Take your time cross-checking in VLC ? the mm:ss values should line up with the seek bar directly. Once you confirm (or correct) the intervals, I'll draft the Roora report covering all of the above and show you the full preview here before producing any PDF.

83. user (2026-06-10T09:48:26.033Z)

I just verified the start and end times, and your 2. is absolutely spot on in that I notice that the start and end of each rally is always off by ~0.5s+ (up to 2s in a few). I want to flag it such that we show the seriousness of the implications of such negligence/incorrectness. Score before/after we can ignore because its coming from their preexisting templates I suppose. But let us crisply ask them to remove **all** the unnecessary and even confusing metrics that they are sharing with us which we have not asked for. You can refer to the v7 of the guidelines and Roora pdfs to understand the details and also we want to highlight the "skipping of dead time often before the match or during etc. and also skipping the warmup rallies. Please also clearly highlight the issue 7. i.e. that rally should not have been counted since they didnt know when it started. Rest of your findings are really great feedbacks except 6 which we should ask to drop because its of no value and unnecessarily polluting. Can you show me a preview of a report based on this? I shall approve and if you give me a pdf then I can respond to them. Unless you want to reuse my Outlook account and find the emails from Jamshad @roora (sent to avidullu@live.com) and formulate a good response which I can then review and press send. Your call

84. assistant (2026-06-10T09:49:13.782Z)

Before drafting, I need the ground truth on what we actually asked Roora for ? the v7 guidelines and their PDFs ? so the "remove everything we didn't ask for" section is precise. Let me locate those:

85. user (2026-06-10T09:49:17.899Z)

C:\Users\avidu\Projects\Annotation Setup
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Annotation Videos
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\annotations
C:\Users\avidu\Projects\sports-data-collector\docs\annotation

86. user (2026-06-10T09:49:17.982Z)

C:\Users\avidu\Projects\sports-data-collector\data\roora_badminton_canonical.csv

C:\Users\avidu\Projects\sports-data-collector\data\roora_badminton_issues.txt

87. user (2026-06-10T09:49:27.515Z)

FullName

Length

C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling
(Supplement).pdf 114515
C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec,
Scope, QA).pdf 80882
C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md
4207
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Badminton
(2026-06-06).pdf 128598

88. user (2026-06-10T09:49:27.596Z)

No files found

89. user (2026-06-10T09:49:38.377Z)

<persisted-output>

Output too large (76.8KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-
Projects-badminton-highlight-indexer\2139f383-5578-4e22-831e-8859ad4f69f2\tool-
results\b2rzrf263.txt

Preview (first 2KB):

C:\Users\avidu\Projects\Annotation Setup\golden_labels_badminton.csv
C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling
(Supplement).pdf
C:\Users\avidu\Projects\Annotation Setup\rally_review.csv
C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec,
Scope, QA).pdf
C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Annotation Videos\Sample Long Video -
Baadminton.MP4
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Annotation Videos\Video 2 - Pickleball.mp4
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Annotation Videos\Video 3 - Tabletennis.mp4
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\badminton\badminton\labelmap.txt
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000000.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000030.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000060.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000090.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000120.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000150.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000180.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000210.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000240.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000270.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000300.xml
...

</persisted-output>

90. user (2026-06-10T09:49:46.389Z)

2:C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling (Supplement).pdf
4:C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec, Scope, QA).pdf
5:C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md
726:C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Rally8ShouldBeThis.png
727:C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Badminton (2026-06-06).pdf
768:C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\roora\annotations.report.md
769:C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\roora\annotations.summary.md

91. user (2026-06-10T09:49:55.118Z)

PDF file read: C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec, Scope, QA).pdf (79KB)

92. user (2026-06-10T09:49:55.698Z)

```
1      # Roora Pilot ? Badminton Review & Conventions for Scale-up
2
3      **Date:** 6 June 2026
4      **Re:** Phase-A calibration pilot ? badminton job (CVAT job id 4075945)
5      **Companion docs:** *Roora Annotation Brief ? Rally Golden Set (v2)*, *Illustrated
Annotation Guide*
6
7      ---
8
9      ## 1. Summary
10
11     Thank you ? the first badminton delivery is a solid calibration round. We've built an
ingestion
12     adapter on our side, so **your CVAT export works for us directly** and the timing/format
mechanics
13     are handled here, not in your workflow. After a full frame-by-frame review, the rally
**timings
14     are generally good**; the items that genuinely need your attention are about
**annotation
15     completeness and rally boundaries** ? and they matter most for the larger batches coming
next.
16
17     ---
18
19     ## 2. Handled on our side ? no change needed from you
20
21     - **Export format.** Keep working in **CVAT**; we ingest your bounding-box export as-is.
No need
22     to produce a CSV.
23     - **Timestamps.** We recompute each rally's time from the box **frame range at the true
fps**
24     (`seconds = frame / fps`), so the CVAT time attributes don't matter, and `mm:ss` vs
decimal is
25     a non-issue ? our ingestion parses both. Minor boundary slack is absorbed by padding
downstream.
26     - **Per-frame label drift.** Where a rally's boxes carried slightly different attribute
values
27     across frames, we collapse to one value per rally automatically.
28
29     ## 3. What needs your attention (for the upcoming larger batches)
30
31     These can't be fixed in code ? they're annotation-quality conventions.
```

```

32
33     ### 3.1 Completeness ? don't miss rallies *(most important)*
34     Our review found rallies with **no label**:
35     - **~1:46** ? a 5-shot rally omitted entirely (frame grab attached).
36     - **3:34?3:43** ? a full rally omitted because it was **absorbed into the previous
rally's box**
37     (the box ran from 3:25 all the way to 3:43, covering two rallies + the dead time
between).
38
39     The brief calls a missed rally "the most serious error." For 30-min videos this is the
#1 thing
40     to get right.
41
42     ### 3.2 One rally = one continuous interval
43     A few boxes spanned **past the shuttle-dead frame** or **across two rallies**:
44
45     | rally | delivered span | true rally | issue |
46     |-----|-----|-----|-----|
47     | 2 | 0:02?0:32 | 0:29?0:32 | box started ~27 s into dead time |
48     | 7 | 1:35?1:51 | 1:35?1:38 | box ran ~13 s past the end |
49     | 16 | 3:25?3:43 | 3:25?3:29 | box covered a **second** rally (see 3.1) |
50
51     Convention: start the box at the **serve-contact** frame, end it at the **shuttle-dead**
frame,
52     and start a **new** box for the next rally ? never let one box bridge a gap.
53
54     ### 3.3 Endings ? `winner` vs an error
55     A `winner` means the opponent made **no legal contact**. A few rallies where the shuttle
landed
56     **out** were marked `winner` (e.g. 6, 7, 9). If the rally ended on a miss, it's a
57     `forced_error`/`unforced_error`, not a winner.
58
59     ### 3.4 New field ? `last_shot_outcome` *(small label-schema add)*
60     We're adding one attribute so the physical outcome is captured alongside the reason:
61
62     | field | values |
63     |-----|-----|
64     | `ending_reason` | winner \| forced_error \| unforced_error \| service_fault \| let \|
other |
65     | `last_shot_outcome` | in \| net \| out \| n_a |
66
67     Just add `last_shot_outcome` to the CVAT label schema (one dropdown). If that's awkward,
tell us
68     and we'll keep it optional.
69
70     ### 3.5 Scope ? pickleball & table tennis
71     The pilot is 1 badminton + 1 pickleball + 1 table tennis; we've received badminton only.
Please
72     confirm status / ETA for the other two.
73
74     ---
75
76     ## 4. Convention checklist (for the next batches)
77
78     - [ ] **Every** served point labeled ? no missed rallies, including back-to-back ones.
79     - [ ] One box per rally: serve-contact ? shuttle-dead; never bridge a gap or two
rallies.
80     - [ ] `ending_reason` ? {winner, forced_error, unforced_error, service_fault, let,
other}.
81     - [ ] `last_shot_outcome` ? {in, net, out, n_a} (one dropdown).
82     - [ ] `shots_count` = total racket contacts (serve = 1).
83     - [ ] CVAT export is fine ? we handle the rest.
84
85     *(Open items on our side to confirm with you: which brief version your team worked from,
and that

```


86 the per-video fps will be in the manifest going forward.)*
87

93. user (2026-06-10T09:51:22.331Z)

PDF file read: C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling (Supplement).pdf (111.8KB)

94. assistant (2026-06-10T09:53:05.332Z)

I now have everything grounded: the Brief and Illustrated Guide on disk are both ****DRAFT v2**** (the badminton review references locking "Guide v1.1") ? I found no "v7" anywhere locally, so I've cited v2 sections in the draft below rather than invent a version; if v7 lives in the email thread, I'll realign the references when I can see it. One more forensic detail that strengthens the boundary complaint: ****all 24 rally boundaries (12 rallies × start/end) in their two files land exactly on multiples of 30 frames**** ? they only ever placed boxes on the 0.5 s sampling grid and never on the true contact/dead frame, which structurally guarantees the 0.5 s error you measured.

Here's the report preview:

Roora Pilot Review ? Pickleball & Table Tennis

****Date:**** 10 June 2026

****Re:**** Phase-A calibration pilot ? pickleball + table tennis deliveries (CVAT exports, `Video 2 - Pickleball.mp4`, `Video 3 - Tabletennis.mp4`)

****Companion docs:**** *Roora Annotation Brief ? Rally Golden Set (v2)*, *Illustrated Annotation Guide (v2)*, *Pilot Review ? Badminton (2026-06-06)*

1. Summary

Thank you for completing the pilot trio ? both deliveries ingest cleanly, shot counts are filled and plausible, numbering and rally separation are clean. However, this round has ****one serious issue that blocks acceptance: rally boundary accuracy.**** Every rally's start/end in both videos is off by ****0.5 s or more ? up to ~2 s in several**** ? against the agreed tolerance of ****±0.3 s**** (Brief §4, Guide Part 3). The badminton round's timings were generally good, so this is a regression; we'd like to understand what changed in the process. Both videos need correction and redelivery (rework per Brief §4) before Phase B can start.

2. Critical ? boundary accuracy misses the agreed acceptance bar

****What we found.**** A full frame-by-frame re-check of all 12 rallies shows every `start`/`end` boundary off by 0.5 s, several by up to ~2 s.

****Root cause (from your files).**** All 24 rally boundaries sit ****exactly on the 30-frame sampling grid**** ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid frames ****cannot**** meet ±0.3 s. Sampling every 30th frame ****inside**** a rally is fine; but the ****first box must be on the exact serve-contact frame and the last box on the exact ball-dead frame**** (Guide Part 3), even when those fall between grid steps.

****Why this matters so much.**** These labels are the golden ground truth we train and evaluate a rally-segmentation model against ? ****boundary precision is the product itself****. The rallies in these videos run 178.5 s, so a 0.5?2 s boundary error corrupts 10?100% of a rally's span; the 1.0 s table-tennis rally #2 can be erased entirely by it. Concretely: at our standard evaluation gate (overlap ? 50%), a typical 4.5 s rally labeled 2 s late ****fails to match even a perfectly correct detection**** ? the dataset would score correct model output as wrong and wrong output as right. Imprecise gold is worse than no data, because it silently corrupts every model trained on it and every quality number reported from it.

****Ask:**** redeliver both videos with frame-accurate first/last boxes, self-checked against ± 0.3 s before submission (Guide Part 7 checklist).

3. Critical ? do not label a rally whose start is not in the video

Pickleball ****rally #1 starts at the very first frame of the file (0:00.000)****. The serve contact is not observable ? the video begins mid-action ? so its start time is unknowable and it cannot meet the boundary bar. The Guide already covers this: **"If unsure whether the match has started, flag it to us rather than guessing"** (Part 6). ****Please remove this rally in the redelivery**** and, going forward, flag any play already in progress at the start of a file instead of labeling it.

4. Reminder for Phase B ? dead time and warm-up

Phase-B videos are ~30 minutes and will contain substantial non-match footage, so we want to re-emphasize two rules now (Guide Rule 1 + Part 6):

- ****Skip all dead time**** ? before the match, between rallies, during breaks: walking, picking up the ball, serve preparation, changeovers, coaching. None of it is ever labeled.
- ****Skip warm-up / knock-up rallies and practice points**** ? they are not real rallies. Labeling starts at the ****first real, scored serve****. When in doubt whether the match has started, flag it to us.

5. Schema hygiene ? remove the fields we did not ask for

Please remove the following from the CVAT label schema; they add noise and, in one case, actively confusing data:

- ****`start time` / `end time`**** ? the delivered values are decimal minutes computed at an assumed 15 fps, ~4x off from real time (the videos are 59.94 fps). We derive all timing from the box frame ranges, so these fields serve no purpose and can only mislead. Remove them.
- ****`score\_before(A-B)` / `score\_after(A-B)`**** ? empty on every box in both videos. We are dropping scores from the schema; remove them.

The schema going forward is exactly: ****`rally\_number`, `shot\_count`, `ending\_reason`, `last\_shot\_outcome`, `note`**** ? nothing else.

6. Field quality

- ****`last\_shot\_outcome`**** ? pickleball has the literal value ``0`` on all 6 rallies (not a valid value); table tennis left it blank. The vocabulary is ``in | net | out | n_a`` ? please use it.
- ****`ending\_reason` = `other` is overused**** ? 5 of 6 table-tennis rallies. ``other`` is the **last resort** (Guide Part 4): "out of table" and "net" endings map to ``forced_error`/`unforced_error`` (with ``last_shot_outcome`` = ``out`/`net``). Also, a note is ****required**** whenever ``other`` is used ? 3 of the 5 ``other`` rallies have none.
- ****One value per rally**** ? table-tennis rally #5 carries ``other`` on some frames and ``winner`` on others. Attribute values must be identical on every box of a rally.

7. What was good ? please keep doing it

- In-rally sampling discipline is excellent: boxes every 30 frames with zero gaps in all 12 rallies.
- Clean sequential numbering, no overlapping rallies.
- ``shot_count`` filled everywhere and plausible against rally durations (including the fast 12-shot table-tennis rally).
- Both sports delivered together, in a format our ingestion reads directly.

8. Redelivery checklist

- ☐ Pickleball + table tennis redelivered with first/last boxes on the exact serve-contact / ball-dead frames (± 0.3 s).
- ☐ Pickleball rally #1 removed (start not observable in the file).

- [] `start time`, `end time`, `score\_before(A-B)`, `score\_after(A-B)` removed from the label schema.
- [] `last\_shot\_outcome` ? `in | net | out | n\_a` on every rally.
- [] `other` only as last resort, always with a note; taxonomy per the Guide Part 4 decision order.
- [] Identical attribute values on every box of a rally.
- [] Please confirm which version of the Brief/Guide your team worked from.

Phase B is gated on a clean redelivery of these two videos.

A couple of notes on choices I made, for your review:

- ****Tone:**** the seriousness comes through their own contract terms ? their brief's §4 acceptance criteria (± 0.3 s, rework in scope) and the "imprecise gold corrupts every model and metric" implication ? rather than words like "negligence." I framed it as a regression vs. their badminton round, which both stings and is fair. Happy to sharpen or soften.
- ****The 30-frame-grid root cause**** is included deliberately: it shows we did forensics they can't hand-wave away, and gives them a precise, actionable fix.
- ****Internal items excluded:**** our converter's `shot\_count` mapping bug (our side, chip already spawned) isn't mentioned to them.
- ****"v7":**** not found on disk (everything local is v2) ? tell me if v7 is something from the email thread and I'll fix the references.

On delivery, my call: once you approve the text, I'll generate the PDF in the same style as the badminton review PDF, and then also draft the reply on the Jamshad thread for you to review and send ? the thread lives on your `avidullu@live.com` Outlook account, which I can reach via the browser if you're signed in (the connected email tooling here appears to be your Gmail). If Outlook access turns out to be awkward, you'll have the PDF either way. Want any edits first?

95. user (2026-06-10T10:09:43.583Z)

I guess you should just give me the pdf and I will word the email appropriately. Also can you remove the redelivery part from the report and emphasize that if the video starts at an ongoing rally then they should skip it and start annotating from the first rally which starts in the video?

96. assistant (2026-06-10T10:10:21.864Z)

I'll revise the report (drop the redelivery section, make "skip rallies already in progress at file start" the headline convention in §3) and produce the PDF.

97. user (2026-06-10T10:10:23.130Z)

Launching skill: anthropic-skills:pdf

98. user (2026-06-10T10:10:23.121Z)

Base directory for this skill: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\skills-plugin\b4180250-5146-4b86-9001-c9f3dd74258c\87f748ce-e72b-44f3-8ef0-78e3212a99c3\skills\pdf

PDF Processing Guide

Overview

This guide covers essential PDF processing operations using Python libraries and command-line tools. For advanced features, JavaScript libraries, and detailed examples, see REFERENCE.md. If you need to fill out a PDF form, read FORMS.md and follow its instructions.

Quick Start

```
```python
from pypdf import PdfReader, PdfWriter
```

## Read a PDF

```
reader = PdfReader("document.pdf")
print(f"Pages: {len(reader.pages)}")
```

## Extract text

```
text = ""
for page in reader.pages:
 text += page.extract_text()
...
```

## Python Libraries

### pypdf - Basic Operations

```
Merge PDFs
```python
from pypdf import PdfWriter, PdfReader

writer = PdfWriter()
for pdf_file in ["doc1.pdf", "doc2.pdf", "doc3.pdf"]:
    reader = PdfReader(pdf_file)
    for page in reader.pages:
        writer.add_page(page)

with open("merged.pdf", "wb") as output:
    writer.write(output)
...
```

```
#### Split PDF
```python
reader = PdfReader("input.pdf")
for i, page in enumerate(reader.pages):
 writer = PdfWriter()
 writer.add_page(page)
 with open(f"page_{i+1}.pdf", "wb") as output:
 writer.write(output)
...
```

```
Extract Metadata
```python
reader = PdfReader("document.pdf")
meta = reader.metadata
print(f"Title: {meta.title}")
print(f"Author: {meta.author}")
print(f"Subject: {meta.subject}")
print(f"Creator: {meta.creator}")
...
```

```
#### Rotate Pages
```python
reader = PdfReader("input.pdf")
writer = PdfWriter()

page = reader.pages[0]
page.rotate(90) # Rotate 90 degrees clockwise
writer.add_page(page)

with open("rotated.pdf", "wb") as output:
 writer.write(output)
...
```

## pdfplumber - Text and Table Extraction

```
Extract Text with Layout
```python
import pdfplumber

with pdfplumber.open("document.pdf") as pdf:
    for page in pdf.pages:
        text = page.extract_text()
        print(text)
...

#### Extract Tables
```python
with pdfplumber.open("document.pdf") as pdf:
 for i, page in enumerate(pdf.pages):
 tables = page.extract_tables()
 for j, table in enumerate(tables):
 print(f"Table {j+1} on page {i+1}:")
 for row in table:
 print(row)
...

Advanced Table Extraction
```python
import pandas as pd

with pdfplumber.open("document.pdf") as pdf:
    all_tables = []
    for page in pdf.pages:
        tables = page.extract_tables()
        for table in tables:
            if table: # Check if table is not empty
                df = pd.DataFrame(table[1:], columns=table[0])
                all_tables.append(df)
```

Combine all tables

```
if all_tables:
    combined_df = pd.concat(all_tables, ignore_index=True)
    combined_df.to_excel("extracted_tables.xlsx", index=False)
...
```

reportlab - Create PDFs

```
#### Basic PDF Creation
```python
from reportlab.lib.pagesizes import letter
from reportlab.pdfgen import canvas

c = canvas.Canvas("hello.pdf", pagesize=letter)
width, height = letter
```

## Add text

```
c.drawString(100, height - 100, "Hello World!")
c.drawString(100, height - 120, "This is a PDF created with reportlab")
```

## Add a line

```
c.line(100, height - 140, 400, height - 140)
```

## Save

```
c.save()
...
```

#### #### Create PDF with Multiple Pages

```
```python
from reportlab.lib.pagesizes import letter
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, PageBreak
from reportlab.lib.styles import getSampleStyleSheet
```

```
doc = SimpleDocTemplate("report.pdf", pagesize=letter)
styles = getSampleStyleSheet()
story = []
```

Add content

```
title = Paragraph("Report Title", styles['Title'])
story.append(title)
story.append(Spacer(1, 12))
```

```
body = Paragraph("This is the body of the report. " * 20, styles['Normal'])
story.append(body)
story.append(PageBreak())
```

Page 2

```
story.append(Paragraph("Page 2", styles['Heading1']))
story.append(Paragraph("Content for page 2", styles['Normal']))
```

Build PDF

```
doc.build(story)
```
```

#### #### Subscripts and Superscripts

**\*\*IMPORTANT\*\***: Never use Unicode subscript/superscript characters (????????, ?<sup>123</sup>????) in ReportLab PDFs. The built-in fonts do not include these glyphs, causing them to render as solid black boxes.

Instead, use ReportLab's XML markup tags in Paragraph objects:

```
```python
from reportlab.platypus import Paragraph
from reportlab.lib.styles import getSampleStyleSheet
```

```
styles = getSampleStyleSheet()
```

Subscripts: use <sub> tag

```
chemical = Paragraph("H<sub>2</sub>O", styles['Normal'])
```

Superscripts: use <sup> tag

```
squared = Paragraph("x<sup>2</sup> + y<sup>2</sup>", styles['Normal'])
```
```

For canvas-drawn text (not Paragraph objects), manually adjust font the size and position rather than using Unicode subscripts/superscripts.

## Command-Line Tools

### pdftotext (poppler-utils)

```
```bash
```

Extract text

```
pdftotext input.pdf output.txt
```

Extract text preserving layout

```
pdftotext -layout input.pdf output.txt
```

Extract specific pages

```
pdftotext -f 1 -l 5 input.pdf output.txt # Pages 1-5
```
```

### qpdf

```
```bash
```

Merge PDFs

```
qpdf --empty --pages file1.pdf file2.pdf -- merged.pdf
```

Split pages

```
qpdf input.pdf --pages . 1-5 -- pages1-5.pdf
qpdf input.pdf --pages . 6-10 -- pages6-10.pdf
```

Rotate pages

```
qpdf input.pdf output.pdf --rotate=+90:1 # Rotate page 1 by 90 degrees
```

Remove password

```
qpdf --password=mypassword --decrypt encrypted.pdf decrypted.pdf
```
```

### pdftk (if available)

```
```bash
```

Merge

```
pdftk file1.pdf file2.pdf cat output merged.pdf
```

Split

```
pdftk input.pdf burst
```

Rotate

```
pdftk input.pdf rotate 1east output rotated.pdf
```
```

## Common Tasks

### Extract Text from Scanned PDFs

```
```python
```

Requires: pip install pytesseract pdf2image

```
import pytesseract
from pdf2image import convert_from_path
```

Convert PDF to images

```
images = convert_from_path('scanned.pdf')
```

OCR each page

```
text = ""
for i, image in enumerate(images):
    text += f"Page {i+1}:\n"
    text += pytesseract.image_to_string(image)
    text += "\n\n"

print(text)
```
```

### Add Watermark

```
```python
from pypdf import PdfReader, PdfWriter
```

Create watermark (or load existing)

```
watermark = PdfReader("watermark.pdf").pages[0]
```

Apply to all pages

```
reader = PdfReader("document.pdf")
writer = PdfWriter()
```

```
for page in reader.pages:
    page.merge_page(watermark)
    writer.add_page(page)
```

```
with open("watermarked.pdf", "wb") as output:
    writer.write(output)
...
```

Extract Images

```
```bash
```

## Using pdftimages (poppler-utils)

```
pdftimages -j input.pdf output_prefix
```

## This extracts all images as output\_prefix-000.jpg, output\_prefix-001.jpg, etc.

```
...
```

## Password Protection

```
```python
```

```
from pypdf import PdfReader, PdfWriter
```

```
reader = PdfReader("input.pdf")
writer = PdfWriter()
```

```
for page in reader.pages:
    writer.add_page(page)
```

Add password

```
writer.encrypt("userpassword", "ownerpassword")
```

```
with open("encrypted.pdf", "wb") as output:
    writer.write(output)
...
```

Quick Reference

Task	Best Tool	Command/Code
Merge PDFs	pypdf	`writer.add_page(page)`
Split PDFs	pypdf	One page per file
Extract text	pdfplumber	`page.extract_text()`
Extract tables	pdfplumber	`page.extract_tables()`
Create PDFs	reportlab	Canvas or Platypus
Command line merge	qpdf	`qpdf --empty --pages ...`
OCR scanned PDFs	pytesseract	Convert to image first
Fill PDF forms	pdf-lib or pypdf	(see FORMS.md) See FORMS.md

Next Steps

- For advanced pypdfium2 usage, see REFERENCE.md
- For JavaScript libraries (pdf-lib), see REFERENCE.md
- If you need to fill out a PDF form, follow the instructions in FORMS.md
- For troubleshooting guides, see REFERENCE.md

ARGUMENTS: Create a professional PDF report (pilot review document for an annotation vendor) from markdown-like content, matching a simple clean document style

99. user (2026-06-10T10:11:13.835Z)

Exit code 1

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
    import reportlab; print(reportlab.Version)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

ModuleNotFoundError: No module named 'reportlab'

100. user (2026-06-10T10:11:28.609Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
4.5.1

101. assistant (2026-06-10T10:12:00.976Z)

Roora Pilot Review ? Pickleball & Table Tennis

Date: 10 June 2026
Re: Phase-A calibration pilot ? pickleball + table tennis deliveries (CVAT exports; `Video 2 - Pickleball.mp4`, `Video 3 - Tabletennis.mp4`)
Companion docs: *Roora Annotation Brief ? Rally Golden Set (v2)*, *Illustrated Annotation Guide (v2)*, *Pilot Review ? Badminton (2026-06-06)*

1. Summary

Thank you for completing the pilot trio ? both deliveries ingest cleanly, shot counts are filled and plausible, and numbering and rally separation are clean. However, this round has **one serious issue: rally boundary accuracy.** Every rally's start/end in both videos is off by **0.5 s or more ? up to ~2 s in several** ? against the agreed tolerance of **±0.3 s** (Brief §4, Guide Part 3). The badminton round's timings were generally good, so this is a regression; we'd like to understand what changed in the process. The sections below explain the impact, the root cause we found in the files, and the conventions to lock before the Phase-B batches.

2. Boundary accuracy ? outside the agreed tolerance ***(most important)***

What we found. A full frame-by-frame re-check of all 12 rallies shows every `start`/`end` boundary off by 0.5 s or more, several by up to ~2 s.

Root cause (from your files). All 24 rally boundaries sit **exactly** on the 30-frame sampling grid ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid frames **cannot** meet ±0.3 s. Sampling every 30th frame **inside** a rally is fine; but the **first** box must be on the exact serve-contact frame and the last box on the exact ball-dead frame (Guide Part 3), even when those fall between grid steps.

Why this matters so much. These labels are the golden ground truth we train and evaluate a rally-segmentation model against ? **boundary precision is the product itself**. The rallies in these videos run 1.8 s, so a 0.5 s boundary error corrupts 100% of a rally's span; the 1.0 s table-tennis rally #2 can be erased entirely by it. Concretely: at our standard evaluation gate (overlap ? 50%), a typical 4.5 s rally labeled 2 s late **fails** to match even a perfectly correct detection ? the dataset would score correct model output as wrong and wrong output as right. Imprecise gold is worse than no data, because it silently corrupts every model trained on it and every quality number reported from it.

Going forward: place the first/last box of every rally on the exact contact/dead frame (frame-step to find it ? Guide Parts 3 and 6), and self-check boundaries against ±0.3 s before submitting (Guide Part 7 checklist).

3. Rallies already in progress when the video starts ? skip them

Pickleball ****rally #1 starts at the very first frame of the file (0:00.000)****. The serve contact is not in the video ? the file begins mid-rally ? so its start time is unknowable and the rally should not have been labeled at all.

****Convention to follow from now on:**** if a video begins during an ongoing rally, ****skip that rally entirely and start annotating from the first rally that starts (serve contact visible) within the video.**** Never extrapolate or guess a boundary you did not see. The same applies at the end of a file: if the video cuts off mid-rally, skip that rally. When in doubt, flag it to us rather than guessing (Guide Part 6).

4. Reminder for Phase B ? dead time and warm-up

Phase-B videos are ~30 minutes and will contain substantial non-match footage, so we want to re-emphasize two rules now (Guide Rule 1 + Part 6):

- ****Skip all dead time**** ? before the match, between rallies, during breaks: walking, picking up the ball, serve preparation, changeovers, coaching. None of it is ever labeled.
- ****Skip warm-up / knock-up rallies and practice points**** ? they are not real rallies. Labeling starts at the ****first real, scored serve****. When in doubt whether the match has started, flag it to us.

5. Schema hygiene ? remove the fields we did not ask for

Please remove the following from the CVAT label schema; they add noise and, in one case, actively confusing data:

- ****`start time` / `end time`**** ? the delivered values are decimal minutes computed at an assumed 15 fps, ~4x off from real time (the videos are 59.94 fps). We derive all timing from the box frame ranges, so these fields serve no purpose and can only mislead. Remove them.
- ****`score\_before(A-B)` / `score\_after(A-B)`**** ? empty on every box in both videos. We are dropping scores from the schema; remove them.

The schema going forward is exactly: ****`rally\_number`, `shot\_count`, `ending\_reason`, `last\_shot\_outcome`, `note`**** ? nothing else.

6. Field quality

- ****`last\_shot\_outcome`**** ? pickleball has the literal value ``0`` on all 6 rallies (not a valid value); table tennis left it blank. The vocabulary is ``in | net | out | n_a`` ? please use it.
- ****`ending\_reason` = `other` is overused**** ? 5 of 6 table-tennis rallies. ``other`` is the **last resort** (Guide Part 4): "out of table" and "net" endings map to ``forced_error`/`unforced_error`` (with ``last_shot_outcome`` = ``out`/`net``). Also, a note is ****required**** whenever ``other`` is used ? 3 of the 5 ``other`` rallies have none.
- ****One value per rally**** ? table-tennis rally #5 carries ``other`` on some frames and ``winner`` on others. Attribute values must be identical on every box of a rally.

7. What was good ? please keep doing it

- In-rally sampling discipline is excellent: boxes every 30 frames with zero gaps in all 12 rallies.
- Clean sequential numbering, no overlapping rallies.
- ``shot_count`` filled everywhere and plausible against rally durations (including the fast 12-shot table-tennis rally).
- Both sports delivered together, in a format our ingestion reads directly.

8. Before Phase B

- We will fold the conventions above (exact first/last frames, skip rallies in progress at file start/end, dead-time and warm-up rules, the reduced field schema) into the next revision of the Guide ? this review is the joint calibration loop the Brief (§3, Phase A) describes.

- Please confirm which version of the Brief/Guide your team worked from on these two videos.

102. user (2026-06-10T10:12:00.998Z)

File created successfully at: C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md (file state is current in your context ? no need to Read it back)

103. assistant (2026-06-10T10:12:12.713Z)

Now the PDF generator script:

104. assistant (2026-06-10T10:12:49.856Z)

```
"""Render the Roora pickleball/table-tennis pilot review as a clean PDF.
```

```
Mirrors 'Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md'.
Glyph note: Helvetica/WinAnsi has no GREATER-THAN-OR-EQUAL ? use '>=' in text.
"""
```

```
from reportlab.lib.pagesizes import letter
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib.units import inch
from reportlab.lib import colors
from reportlab.platypus import (
    SimpleDocTemplate, Paragraph, Spacer, HRFflowable, ListFlowable, ListItem,
)
```

```
OUT = r"C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review -
Pickleball and Table Tennis (2026-06-10).pdf"
```

```
ss = getSampleStyleSheet()
DARK = colors.HexColor("#1a1a1a")
GREY = colors.HexColor("#444444")
```

```
title = ParagraphStyle("title", parent=ss["Title"], fontName="Helvetica-Bold",
                        fontSize=17, leading=21, alignment=0, spaceAfter=10,
                        textColor=DARK)
meta = ParagraphStyle("meta", parent=ss["Normal"], fontName="Helvetica",
                      fontSize=9.5, leading=13.5, textColor=GREY)
h2 = ParagraphStyle("h2", parent=ss["Heading2"], fontName="Helvetica-Bold",
                   fontSize=12.5, leading=15, spaceBefore=14, spaceAfter=5,
                   textColor=DARK)
body = ParagraphStyle("body", parent=ss["Normal"], fontName="Helvetica",
                     fontSize=10, leading=14, spaceAfter=6, textColor=DARK)
bullet = ParagraphStyle("bullet", parent=body, leftIndent=2, spaceAfter=4)
```

```
CODE = '<font face="Courier" size="9.2">%s</font>'
```

```
def P(text, style=body):
    return Paragraph(text, style)
```

```
def UL(items):
    return ListFlowable(
        [ListItem(P(t, bullet), leftIndent=16, value="circle") for t in items],
        bulletType="bullet", start="circle", bulletFontSize=6,
        leftIndent=14, spaceAfter=6,
    )
```

```
story = [
    P("Roora Pilot Review ? Pickleball & Table Tennis", title),
    P("<b>Date:</b> 10 June 2026", meta),
```

P("Re: Phase-A calibration pilot ? pickleball + table tennis deliveries " " (CVAT exports; " + CODE % "Video 2 - Pickleball.mp4" + ", " + CODE % "Video 3 - Tabletennis.mp4" + ") ", meta), P("Companion docs: <i>Roora Annotation Brief ? Rally Golden Set (v2)</i>, " "<i>Illustrated Annotation Guide (v2)</i>, <i>Pilot Review ? Badminton (2026-06-06)</i>", meta), Spacer(1, 8), HRFlowable(width="100%", thickness=0.8, color=colors.HexColor("#999999")),

P("1. Summary", h2), P("Thank you for completing the pilot trio ? both deliveries ingest cleanly, shot counts are "

"filled and plausible, and numbering and rally separation are clean. However, this round has "

"one serious issue: rally boundary accuracy. Every rally's start/end in both videos is "

"off by 0.5 s or more ? up to ~2 s in several ? against the agreed tolerance of "

" ± 0.3 s (Brief §4, Guide Part 3). The badminton round's timings were generally good, "

"so this is a regression; we'd like to understand what changed in the process. The sections "

"below explain the impact, the root cause we found in the files, and the conventions to lock "

"before the Phase-B batches."),

P("2. Boundary accuracy ? outside the agreed tolerance <i>(most important)</i>", h2), P("What we found. A full frame-by-frame re-check of all 12 rallies shows every " + CODE % "start" + "/" + CODE % "end" + " boundary off by 0.5 s or more, several by up to ~2 s."),

P("Root cause (from your files). All 24 rally boundaries sit exactly on the 30-frame "

"sampling grid ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid "

"frames <i>cannot</i> meet ± 0.3 s. Sampling every 30th frame inside a rally is fine; but "

"the first box must be on the exact serve-contact frame and the last box on the exact "

"ball-dead frame (Guide Part 3), even when those fall between grid steps."),

P("Why this matters so much. These labels are the golden ground truth we train and "

"evaluate a rally-segmentation model against ? boundary precision is the product itself.", "

"The rallies in these videos run 178.5 s, so a 0.5?2 s boundary error corrupts 10?100% of a "

"rally's span; the 1.0 s table-tennis rally #2 can be erased entirely by it. Concretely: at our "

"standard evaluation gate (overlap of at least 50%), a typical 4.5 s rally labeled 2 s late "

"fails to match even a perfectly correct detection ? the dataset would score correct "

"model output as wrong and wrong output as right. Imprecise gold is worse than no data, "

"because it silently corrupts every model trained on it and every quality number reported "

"from it."),

P("Going forward: place the first/last box of every rally on the exact contact/dead frame "

"(frame-step to find it ? Guide Parts 3 and 6), and self-check boundaries against ± 0.3 s "

"before submitting (Guide Part 7 checklist)."),

P("3. Rallies already in progress when the video starts ? skip them", h2), P("Pickleball rally #1 starts at the very first frame of the file (0:00.000). The serve "

"contact is not in the video ? the file begins mid-rally ? so its start time is unknowable and "

"the rally should not have been labeled at all."),

P("Convention to follow from now on: if a video begins during an ongoing rally, skip "

"that rally entirely and start annotating from the first rally that starts (serve contact
 "visible) within the video. Never extrapolate or guess a boundary you did not see. The
 "same applies at the end of a file: if the video cuts off mid-rally, skip that rally. When
 in "doubt, flag it to us rather than guessing (Guide Part 6)."),

P("4. Reminder for Phase B ? dead time and warm-up", h2),
 P("Phase-B videos are ~30 minutes and will contain substantial non-match footage, so we want
 to "re-emphasize two rules now (Guide Rule 1 + Part 6):"),
 UL([
 " Skip all dead time ? before the match, between rallies, during breaks: walking,
 "picking up the ball, serve preparation, changeovers, coaching. None of it is ever
 labeled.",
 " Skip warm-up / knock-up rallies and practice points ? they are not real rallies.
 "Labeling starts at the first real, scored serve. When in doubt whether the match
 "has started, flag it to us.",
]),

P("5. Schema hygiene ? remove the fields we did not ask for", h2),
 P("Please remove the following from the CVAT label schema; they add noise and, in one case,
 "actively confusing data:"),
 UL([
 " " + CODE % "start time" + " / " + CODE % "end time" + " ? the delivered values
 are "decimal minutes computed at an assumed 15 fps, ~4x off from real time (the videos are "
 "59.94 fps). We derive all timing from the box frame ranges, so these fields serve no "
 "purpose and can only mislead. Remove them.",
 " " + CODE % "score_before(A-B)" + " / " + CODE % "score_after(A-B)" + " ? empty
 on "every box in both videos. We are dropping scores from the schema; remove them.",
]),
 P("The schema going forward is exactly: " + CODE % "rally_number" + ", "
 + CODE % "shot_count" + ", " + CODE % "ending_reason" + ", "
 + CODE % "last_shot_outcome" + ", " + CODE % "note" + " ? nothing else."),

P("6. Field quality", h2),
 UL([
 " " + CODE % "last_shot_outcome" + " ? pickleball has the literal value "
 + CODE % "0" + " on all 6 rallies (not a valid value); table tennis left it blank. The "
 "vocabulary is " + CODE % "in | net | out | n_a" + " ? please use it.",
 " " + CODE % "ending_reason" + " = " + CODE % "other" + " is overused ? 5 of 6 "
 "table-tennis rallies. " + CODE % "other" + " is the <i>last resort</i> (Guide Part 4):
 " ?out of table? and ?net? endings map to " + CODE % "forced_error" + "/"
 + CODE % "unforced_error" + " (with " + CODE % "last_shot_outcome" + " = " + CODE %
 "out" + "/" + CODE % "net" + "). Also, a note is required whenever " + CODE % "other"
 + " is used ? 3 of the 5 " + CODE % "other" + " rallies have none.",
 " One value per rally ? table-tennis rally #5 carries " + CODE % "other" + " on
 some "frames and " + CODE % "winner" + " on others. Attribute values must be identical on
 every "box of a rally.",
]),

P("7. What was good ? please keep doing it", h2),
 UL([
 "In-rally sampling discipline is excellent: boxes every 30 frames with zero gaps in all

```

12 rallies.",
    "Clean sequential numbering, no overlapping rallies.",
    CODE % "shot_count" + " filled everywhere and plausible against rally durations
(including "
    "the fast 12-shot table-tennis rally).",
    "Both sports delivered together, in a format our ingestion reads directly.",
    ]),

P("8. Before Phase B", h2),
UL([
    "We will fold the conventions above (exact first/last frames, skip rallies in progress
at "
    "file start/end, dead-time and warm-up rules, the reduced field schema) into the next "
    "revision of the Guide ? this review is the joint calibration loop the Brief (§3, Phase
A) "
    "describes.",
    "Please confirm which version of the Brief/Guide your team worked from on these two
videos.",
    ]),
]

doc = SimpleDocTemplate(OUT, pagesize=letter,
                        leftMargin=0.9 * inch, rightMargin=0.9 * inch,
                        topMargin=0.8 * inch, bottomMargin=0.8 * inch,
                        title="Roora Pilot Review - Pickleball and Table Tennis (2026-06-10)",
                        author="Avi Dullu")

doc.build(story)
print("WROTE", OUT)

```

105. user (2026-06-10T10:12:49.876Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\roora\make_review_pdf.py (file state is current in your context ? no need to Read it back)

106. user (2026-06-10T10:12:59.664Z)

WROTE C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf
pages: 2
chars: 5914
odd glyphs: {'?'}

107. user (2026-06-10T10:13:08.769Z)

PDF file read: C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf (7.4KB)

108. user (2026-06-10T10:13:37.681Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

109. assistant (2026-06-10T10:13:24.588Z)

The PDF renders cleanly ? 2 pages in the same simple document style as the badminton review. Recording this checkpoint in memory, then handing over:

110. user (2026-06-10T10:13:46.410Z)

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5      node_type: memory

```

```
6         type: project
7         originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8         ---
9
10        # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12        **Checkpoint:** 2026-06-10 (REPO-TOPOLGY decision panel + C13 plan ? AWAITING OWNER
RATIFICATION). Owner leaning
```

111. user (2026-06-10T10:14:06.668Z)

The file c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

112. assistant (2026-06-10T10:14:22.232Z)

****Your PDF is ready:**** [Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf](C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf) ? 2 pages, same clean style and location as the badminton review PDF. A markdown record copy sits at the `Annotation Setup` root next to the badminton one.

What changed from the preview you saw:

- ****All redelivery asks removed**** ? the §8 checklist is gone, and the "redeliver both videos / remove this rally in the redelivery / Phase B gated on redelivery" phrasings in the summary and §2?3 are reworded as forward-looking conventions. How hard to push on rework is now entirely yours to decide in the email.
- ****§3 is now the convention you asked for, stated as the headline:**** if a video begins during an ongoing rally, skip that rally entirely and start annotating from the first rally whose serve contact is visible within the video ? never extrapolate a boundary you didn't see. I extended it to the symmetric case (video cutting off mid-rally at the end) since it's the same principle; easy to trim if you'd rather not add it.
- §8 is now a short "Before Phase B" note: we'll fold these conventions into the next Guide revision, plus the ask for them to confirm which Brief/Guide version they worked from (relevant given the v7-vs-v2 question ? everything I have locally is DRAFT v2, so the report cites v2).

Everything else is as you approved: the boundary-accuracy section leads with the ± 0.3 s acceptance bar from their own brief, the 30-frame-grid root cause, and the "imprecise gold corrupts every model and metric" implication; dead-time and warm-up skipping get their own Phase-B reminder section; the schema is cut to exactly `rally\_number, shot\_count, ending\_reason, last\_shot\_outcome, note`; and the good practices (complete sampling, clean numbering, filled shot counts) are acknowledged.

If any wording feels off once you're drafting the email around it, tell me and I'll regenerate the PDF in seconds (the generator script is saved). The internal `shot\_count` converter bug stays our side ? the fix chip for the collector repo is pending whenever you want to run it.

113. user (2026-06-10T10:16:01.784Z)

Beautiful! Another window is actively working on the code right now so I will wait for it to complete and then run the collector fix! Appreciate the help!

114. assistant (2026-06-10T10:16:14.346Z)

Sounds good! The chip will be waiting whenever the other window wraps up ? it's self-contained (repro paths, expected counts, and the fix scope are all in the prompt), so it'll run cold in a fresh worktree without needing anything from this conversation.

Recap of where today landed: master fixed and guarded by CI ([PR #63](https://github.com/avidullu/badminton-highlight-indexer/pull/63) merged, both jobs green), the Roora cross-check data verified by you, and the [pilot review PDF](C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review -

Pickleball and Table Tennis (2026-06-10).pdf) ready for your email to Jamshad. The memory checkpoint reflects all of it, so any future session picks up cleanly.

Good luck with the email ? curious what Roora says about that 15 fps assumption. ?